

Theory question bank

Filled in per cluster as I study. Format: question -> my short answer -> (gap?).

Cluster 1 — Foundations / lattice / PDE

Lecture 01

1. Why does a course built around the COS method open with equity options, given that equities are a tiny fraction of total notional outstanding? **A.** Because the European equity call/put is the canonical optionality problem, and every later extension — Heston, jumps, barriers, American/Bermudan — is built on exactly the same scaffolding (risk-neutral discounted expectation of a payoff). Notional is dominated by rates, but the *pricing theory* of optionality is cleanest and most transferable starting from a single stock under GBM, which is also where COS is first derived.
2. Derive the continuously-compounded bank-account value $M(t)$ from the differential argument $dM = rM dt$, and state the discount factor for one euro at time T . **A.** Interest credited is proportional to balance, rate and time: $dM = rM dt$, so $dM/M = r dt$; integrating from 0 to t gives $\ln M(t) - \ln M(0) = rt$, hence $M(t) = M(0)e^{rt}$. With $M(0)=1$ the numéraire is e^{rt} , and one euro at T is worth $D(t,T) = e^{-r(T-t)}$ at time t .
3. State covered interest-rate parity $(1+i_s) = (F_t/S_t)(1+i_c)$ and prove it by a static no-arbitrage replication. What is the “FX basis” and why is it needed in practice? **A.** One domestic unit, two riskless routes over the same horizon: (A) invest domestically $\rightarrow 1+i_c$; (B) convert at spot S_t , invest abroad at i_s , lock reconversion at forward $F_t \rightarrow (F_t/S_t)(1+i_s)$. No-arbitrage forces equality, giving $(1+i_s) = (F_t/S_t)(1+i_c)$ — i.e. the forward is pinned by spot and the two funding rates, model-free. Empirically CIP fails slightly; the FX basis is the spread added to one leg to restore equality, treated as calibrated market data fed into OTC FX pricing.
4. Write the terminal payoffs of a European call and put. Where are they non-smooth, and why does that kink cause numerical trouble for Fourier/COS methods? **A.** $V_C = \max(S_T - K, 0)$, $V_P = \max(K - S_T, 0)$: continuous, piecewise-linear, with a kink at $S=K$. Expanding the *payoff* itself in a Fourier/cosine series rings (Gibbs) at the kink, so the cosine coefficients decay only algebraically; COS sidesteps this by expanding the smooth *density* and integrating the payoff against cosines analytically (the χ_k, ψ_k closed forms), so the kink never enters a series.
5. In the one-step binomial hedge of a short call, derive $\Delta = 20/(S_{up} - S_{dn})$ and then the premium $V_{c,0}$. Compute both for $\{120, 80\}$ and $\{120, 75\}$. Why is case B more expensive? **A.** Hedged portfolio $\Pi = V_{c,0} - \Delta S_0$; set $\Pi_{up} = \Pi_{dn}$ so $\Delta S_{up} - 20 = \Delta S_{dn}$ (with $K=100$, up-payoff 20), giving $\Delta = 20/(S_{up} - S_{dn})$. Imposing $\Pi=0$ gives $V_{c,0} = \Delta(S_0 - S_{dn})$. Case A $\{120, 80\}$: $\Delta = 1/2$, $V = 1/2 \cdot 20 = 10$. Case B $\{120, 75\}$: $\Delta = 4/9$, $V = 4/9 \cdot 25 = 11.11$. Case B is dearer because the wider outcome spread (more volatility) is worth more to a convex, floored payoff.
6. Show that the one-step replication price equals $E^Q[\text{payoff}]$ under the risk-neutral measure, and that the real-world probabilities $p=0.7$, $p=0.4$ never enter. Define q explicitly. **A.** Define $q = (S_0 - S_{dn})/(S_{up} - S_{dn})$ so that $E^Q[S_T] = qS_{up} + (1-q)S_{dn} = S_0$ (martingale under Q , $r=0$). Then $V_{c,0} = \Delta(S_0 - S_{dn}) = q(S_{up} - K) + (1-q) \cdot 0 = E^Q[(S_T - K)^+]$. The price is built purely from the no-arbitrage hedge; the physical p (0.7, 0.4) never appears — only q does.
7. At the fair price, does a perfectly hedged writer make money? Explain $\Pi_{up} = \Pi_{dn} = 0$ and where real-world profit actually comes from. **A.** No. At the replication price the perfectly hedged terminal portfolio is zero in *both* states ($\Pi_{up} = \Pi_{dn} = 0$): no risk, no profit — the fair price is exactly the cost of replication. Profit only arises from selling above replication value (the surplus invested at r), and in reality from bid-ask spread, fees, discrete-rehedging/transaction-cost edge, and bearing vol/jump risk outside the model.
8. Derive put-call parity from two portfolios held to maturity. Which assumptions about the dynamics of S did you use? **A.** $\Pi_A = \text{call} + K \text{ cash}$, $\Pi_B = \text{put} + \text{share}$. At T both equal $\max(S_T, K)$ (using $\max(S_T - K, 0) + K$ and $\max(K - S_T, 0) + S_T$). Equal payoffs in every state $\Rightarrow$ equal value today: $C + Ke^{-r(T-t)} = P + S_t$. No dynamics assumed — only no-arbitrage and a constant rate — so parity is model-free.

9. Prove the lower bound $C(t) \geq \max(S(t) - K e^{-r(T-t)}, 0)$ and use it to argue an American call on a non-dividend stock is never exercised early. Why does the argument fail for puts? **A.** The portfolio (long call + bond worth $Ke^{-r(T-t)}$) pays $\max(S_T, K) \geq S_T$ at T , so it costs $\geq S(t)$ today: $C(t) \geq S(t) - Ke^{-r(T-t)}$, and ≥ 0 , hence the bound. For $r > 0$ this exceeds intrinsic $S(t) - K$, so the live call beats immediate exercise and American = European. For the put, parity gives $P^{\text{eur}} = Ke^{-r(T-t)} - S^{\text{eur}}$ and $Ke^{-r(T-t)} < K$, so the live put can be below intrinsic $K - S$ — banking K early to earn interest can win, so early exercise is genuinely optimal.
10. Apply Ito's lemma to $g = \ln S$ under GBM and obtain $d(\ln S) = (\mu - \sigma^2/2) dt + \sigma dW$. Where does the $-1/2 \sigma^2$ come from, and what goes wrong if you forget it? **A.** With $g = \ln S$, $g_S = 1/S$, $g_{SS} = -1/S^2$: $d \ln S = (\mu S \cdot 1/S + \frac{1}{2} \sigma^2 S^2 \cdot (-1/S^2))dt + \sigma S \cdot (1/S)dW = (\mu - \frac{1}{2} \sigma^2)dt + \sigma dW$. The $-\frac{1}{2} \sigma^2$ is the Itô correction, born of $(dW)^2 = dt$ in the second-order term. Forgetting it biases the simulated/expected stock high: it is exactly the convexity adjustment that keeps $E[S(t)] = S_0 e^{\mu t}$.
11. From the log-price ABM, write the closed-form $S(t)$ and verify $E[S(t)] = S_0 e^{\mu t}$ using the Gaussian MGF. State the distribution of $\ln(S(T)/S_0)$ under Q . **A.** Integrating $d \ln S$ gives $S(t) = S_0 \exp[(\mu - \frac{1}{2} \sigma^2)t + \sigma W(t)]$. Using $E[e^{\sigma W(t)}] = e^{\sigma^2 t/2}$: $E[S(t)] = S_0 e^{(\mu - \frac{1}{2} \sigma^2)t} \cdot e^{\sigma^2 t/2} = S_0 e^{\mu t}$. Under Q ($\mu \rightarrow r$), $\ln(S(T)/S_0) \sim N((r - \frac{1}{2} \sigma^2)T, \sigma^2 T)$ — lognormal stock.
12. State the four defining properties of a Wiener process. What are $E[dW]$, $E[(dW)^2]$, and the quadratic variation $[W]_t$, and why does that force Ito calculus instead of ordinary calculus? **A.** (i) $W(0) = 0$; (ii) independent, stationary increments; (iii) $W(t) \sim N(0, t)$; (iv) a.s. continuous paths. So $E[dW] = 0$, $E[(dW)^2] = dt$, and $[W]_t = t$ (finite quadratic variation, unbounded total variation). Because $(dW)^2 = dt$ contributes at first order in dt , second-order Taylor terms survive — ordinary calculus fails and Itô's correction $\frac{1}{2} \sigma^2 g_{xx} dt$ is needed.
13. Define the Markov property via $P(S_{t+1} \leq s \mid S_t) = P(S_{t+1} \leq s \mid S_t, S_{t-1}, \dots)$. How does it relate to the weak-form EMH, and why does i.i.d. returns imply weak-form EMH but not conversely? **A.** Markov: the one-step-ahead conditional law given the present equals that given the whole past — history beyond S_t carries no extra information, which is precisely weak-form efficiency (prices reflect past prices). i.i.d. returns make the *entire* history useless, which certainly gives Markov/weak-form; but weak-form only requires the one-step conditional law to be unchanged by deeper history, a strictly weaker condition than full independence and identical distribution — so the converse fails.
14. Contrast ABM, GBM and OU: which can go negative, which is multiplicative, which mean-reverts, and what single property do all three share? Which one is the template for the Heston variance process? **A.** ABM ($dX = \mu dt + \sigma dW$): additive noise, Gaussian, can go negative. GBM ($dX = \mu X dt + \sigma X dW$): multiplicative, stays positive, lognormal. OU ($dX = \kappa(\theta - X)dt + \sigma dW$): mean-reverts to θ at speed κ , Gaussian, can go negative. Shared property: all are Markov diffusions driven by Brownian motion. OU is the template for the CIR/Heston variance $dv = \kappa(\theta - v)dt + \gamma \sqrt{v} dW$ — same mean-reverting drift but a $\sqrt{v}$ diffusion that keeps $v \geq 0$.
15. What stylised fact does the SPX QQ-plot reveal (sample quantiles $[-3, 3]$ vs theoretical $[-2.3, 2.3]$), and which later models in the course are introduced to fix it? **A.** Empirical return quantiles ($\approx \pm 3$) are wider than the normal's ($\approx \pm 2.3$): returns have *fatter tails* than Gaussian (large moves occur more often). This is the central failure of GBM and motivates Heston stochastic volatility (L7) and jump-diffusion models (Merton/Kou, L8), which put weight back in the tails and reproduce the implied-vol smile.

Lecture 02

1. Derive Ito's lemma heuristically: why does the second-order term in $g(t, X)$ survive while the others vanish? **A.** Taylor-expand $dg = g_t dt + g_x dX + \frac{1}{2} g_{xx} (dX)^2 + \dots$ With $dX = \mu dt + \sigma dW$, $(dX)^2 = \sigma^2 (dW)^2 + 2\mu\sigma dt dW + \mu^2 (dt)^2$. Under the table $(dW)^2 = dt$, $dW dt = 0$, $(dt)^2 = 0$, only $\sigma^2 (dW)^2 = \sigma^2 dt$ survives at order dt . So $dY = (g_t + \mu g_x + \frac{1}{2} \sigma^2 g_{xx})dt + \sigma g_x dW$: the second-order term survives precisely because Brownian motion has finite quadratic variation $((dW)^2 = dt)$, while genuinely higher-order terms vanish.
2. Apply Ito to $g = \ln S$ under $dS = \mu S dt + \sigma S dW$. Where does the $-\sigma^2/2$ come from, and what is its sign relation to Jensen? **A.** $g_S = 1/S$, $g_{SS} = -1/S^2$: $d \ln S = (\mu - \frac{1}{2} \sigma^2)dt + \sigma dW$.

The $-\frac{1}{2}\sigma^2$ is the Itô term from $\frac{1}{2}g_{SS}(dS)^2 = -\frac{1}{2}\sigma^2 dt$. It is Jensen's inequality made precise: $\ln$ is concave, so $E[\ln S_t]$ grows at the lower rate $\mu - \frac{1}{2}\sigma^2$ while $E[S_t]$ grows at μ ; the gap is exactly $\frac{1}{2}\sigma^2$.

3. Walk through the delta-hedging derivation of the Black-Scholes PDE. At which exact step is $\Delta = V_S$ forced, and why? **A.** Form $\Pi = V - \Delta S$, self-financing so $d\Pi = dV - \Delta dS$. Itô gives $dV = (V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS})dt + \sigma S V_S dW$. Then $d\Pi = (\dots)dt + (\sigma S V_S - \Delta) dS$. The portfolio is locally riskless iff the dW term vanishes, which forces $\Delta = V_S$. Imposing $d\Pi = r\Pi dt$ then yields $V_t + rS V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0$.
4. In that derivation, μ appears in dV but not in the final PDE. Show precisely the cancellation that removes it. **A.** In $d\Pi$ the drift is $V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - \Delta \mu S$. Once $\Delta = V_S$ is substituted, the term $\mu S V_S$ (from dV) and $-\Delta \mu S = -\mu S V_S$ (from $-\Delta dS$) cancel exactly, removing μ entirely. Economically: hedging eliminates exposure to the real-world drift, so only σ and r remain.
5. State the BS PDE, its order/type, and its terminal condition. Why is it solved backward in time? **A.** $V_t + rS V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0$: second-order, linear, parabolic, with terminal data $V(S, T) = \text{payoff } g(S)$. It is solved backward because the only data we know is at maturity (the payoff); discounted expectation propagates value from T back to t , equivalently the substitution $\tau = T - t$ turns it into a forward initial-value problem.
6. Define a risk-neutral measure Q . Starting from $dS = \mu S dt + \sigma S dW^Q$, prove that the martingale property of $e^{-rt} S_t$ forces $\mu' = r$. **A.** Q is a measure equivalent to P under which the discounted asset $e^{-rt} S_t$ is a martingale. By the product rule $d(e^{-rt} S_t) = (\mu' - r)e^{-rt} S_t dt + \sigma e^{-rt} S_t dW^Q$. The stochastic integral is a martingale; the drift dt -integral is a martingale only if it vanishes, i.e. $\mu' - r = 0$. Hence $\mu' = r$ and $dS = rS dt + \sigma S dW^Q$.
7. Under the change from P to Q , the drift goes $\mu \rightarrow r$ but σ is unchanged. Why must σ be preserved (think equivalence / quadratic variation)? **A.** Girsanov shifts only the drift; the diffusion coefficient is invariant. Equivalence of P and Q means they agree on which events have probability zero, and σ is fixed by the path's quadratic variation $[S]_t = \int_0^t \sigma^2 S^2 ds$ — an almost-sure pathwise property. Changing σ would change the set of possible paths' quadratic variation, breaking equivalence; so only the drift can move.
8. State Feynman-Kac and sketch its proof. What is the role of the discounted process Y_s and where is the PDE used? **A.** If V solves $V_t + bV_x + \frac{1}{2}\sigma^2 V_{xx} - rV = 0$ with $V(x, T) = g$, and $dX = b dt + \sigma dW$ with $X_t = x$, then $V(x, t) = E[e^{-\int_t^T r du} g(X_T) | X_t = x]$. Proof: set $Y_s = e^{-\int_t^s r du} V(X_s, s)$; Itô gives $dY_s = (\text{discount}) \cdot [V_t + bV_x + \frac{1}{2}\sigma^2 V_{xx} - rV] ds + (\dots)dW$. The PDE makes the ds -drift vanish, so Y_s is a martingale; taking E of $Y_T - Y_t$ with $Y_t = V(x, t)$ gives the formula.
9. Explain how Feynman-Kac shows the PDE approach and the martingale approach give the same price. State the equivalence "drift = 0 $\Leftrightarrow$ discounted price is a Q -martingale". **A.** The discounted option price $e^{-rt} V$ is a Q -martingale iff its Itô drift $e^{-rt} (V_t + rS V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV)$ is zero — which is exactly the BS PDE. So "PDE holds" $\Leftrightarrow$ "discounted price is a Q -martingale", and applying the martingale property between t and T gives $V = e^{-r(T-t)} E^Q[g(S_T)]$. PDE route and expectation route are two faces of the same identity.
10. Derive $C_0 = e^{-rT} E^Q[(S_T - K)^+]$ down to $S_0 N(d_1) - K e^{-rT} N(d_2)$. What measure-change / square-completion produces the shift from d_2 to d_1 ? **A.** Write $C_0 = e^{-rT} \int_{-\infty}^{\infty} \ln(K/S_0) \phi(S_0 e^z - K) f_Z(z) dz$ with $Z \sim N((r - \frac{1}{2}\sigma^2)T, \sigma^2 T)$. The K -term is $K e^{-rT} \cdot Q(S_T > K) = K e^{-rT} N(d_2)$. For the $S_0 e^z$ -term, completing the square in $z - (z - m)^2 / 2v^2$ gives $S_0 e^z f_Z(z) = S_0 e^{rT} \cdot (N(m + v^2, v^2) \text{ density})$, i.e. a Girsanov shift of the mean up by $v^2 = \sigma^2 T$ (change to the stock numéraire). That shift turns d_2 into $d_1 = d_2 + \sigma\sqrt{T}$, giving $C_0 = S_0 N(d_1) - K e^{-rT} N(d_2)$.
11. Interpret $N(d_2)$ and $N(d_1)$ probabilistically. Which one equals $Q(S_T > K)$? **A.** $N(d_2) = Q(S_T > K)$, the risk-neutral probability of finishing in the money. $N(d_1)$ is the same exercise probability but under the *stock-numéraire* measure Q^S ; equivalently $S_0 N(d_1)$ is the present value (in numéraire terms) of receiving the stock conditional on exercise. So $Q(S_T > K) = N(d_2)$.
12. Derive put-call parity from a static replication argument (not from the BS formula). Why is parity model-free? **A.** Long one call, short one put (same K, T) pays $(S_T - K)^+ - (K - S_T)^+ = S_T - K$ in every state. A payoff $S_T - K$ at T is replicated today by one share and short K bonds, value $S_0 - K e^{-rT}$. No-arbitrage $\Rightarrow C_0 - P_0 = S_0 - K e^{-rT}$. It is model-free because it uses only the algebraic payoff identity, a static portfolio, and a constant rate — no assumption on the

dynamics of S .

13. Greeks: define Delta, Gamma, Theta. Show the BS PDE as an algebraic identity linking them, and give the closed-form call Delta and Gamma. **A.** $\Delta = \partial V / \partial S$, $\Gamma = \partial^2 V / \partial S^2$, $\Theta = \partial V / \partial t$. The BS PDE is literally $\Theta + rS\Delta + \frac{1}{2}\sigma^2 S^2 \Gamma = rV$ — an algebraic relation tying the Greeks together (they are not independent). For the call $\Delta = N(d_1) \in (0,1)$ and $\Gamma = N'(d_1)/(S_0 \sigma \sqrt{T}) > 0$.
14. List the five parameters the BS call price depends on and explain why μ (the real-world drift) is not among them. **A.** The call depends on S , K , T (or τ), r , σ . The physical drift μ is absent because delta-hedging removes exposure to it (the $\mu S V_S$ terms cancel), equivalently because under Q the drift is replaced by r . Risk preferences are hedged away; only the cost of carry r and the volatility σ matter.

Lecture 03

1. In the implied-volatility problem, which four inputs of the BS price are known and which one is the unknown? Why is recovering σ a root-find rather than a closed-form inversion? **A.** Known: τ , K , r , S_t (all observable). Unknown: σ . Recovering σ from C^{mkt} is a root-find of $F(\sigma) = C^{\text{BS}(\sigma)} - C^{\text{mkt}} = 0$ because σ sits inside the nonlinear $N(d_1)$, $N(d_2)$; there is no algebraic inversion of the normal CDF, so we iterate.
2. Prove vega $= \partial C / \partial \sigma = S_0 \sqrt{T} \phi(d_1) > 0$ for $\sigma > 0$. Why does strict monotonicity of price in σ guarantee a unique implied vol? **A.** Differentiating $C = S_t N(d_1) - Ke^{-r\tau} N(d_2)$ and using the identity $S_t N'(d_1) = Ke^{-r\tau} N'(d_2)$ (from $N'(d_1)/N'(d_2) = e^{\{-\ln(S/K) - r\tau\}}$ since $d_2 = d_1 - \sigma\sqrt{\tau}$), the $\partial d_1 / \partial \sigma$ terms cancel and Vega $= Ke^{-r\tau} N'(d_2) \sqrt{\tau} = S_t \sqrt{\tau} N'(d_1) > 0$. Strict positivity $\Rightarrow C(\sigma)$ is strictly increasing, hence a continuous bijection onto its range, so at most one σ solves $F(\sigma) = 0$.
3. State the admissible no-arbitrage price interval for a European call. What happens to the implied vol as the market price approaches the upper/lower bound? **A.** As $\sigma \rightarrow 0^+$, $C \rightarrow \max(S_t - Ke^{-r\tau}, 0)$ (intrinsic); as $\sigma \rightarrow \infty$, $C \rightarrow S_t$. So C is a bijection from $[0, \infty)$ onto $[\max(S_t - Ke^{-r\tau}, 0), S_t]$. A quote at the lower bound forces $\sigma^{\text{impl}} \rightarrow 0$; a quote approaching the upper bound S_t forces $\sigma^{\text{impl}} \rightarrow \infty$. A quote outside the band has no real implied vol (static-arbitrage violation / stale quote).
4. Bisection vs Newton-Raphson for IV inversion: state the convergence order of each and Newton's failure mode. Why is dividing by vega dangerous deep OTM / short maturity? **A.** Bisection: linear, rate $\frac{1}{2}$ (one bit/step), globally fail-safe given a sign-changing bracket. Newton $\sigma_{n+1} = \sigma_n - F(\sigma_n) / \text{Vega}(\sigma_n)$: quadratic but only local — a poor start can overshoot, diverge, or step to negative σ . Vega $= S_t \sqrt{\tau} N'(d_1)$ is a Gaussian bump that $\rightarrow 0$ deep ITM/OTM and at short τ ; dividing by a near-zero vega makes the Newton step explode, so the inversion is ill-conditioned in the wings.
5. Globalised Newton: why seed at the price-curve inflection point $\hat{\sigma}$? What property there makes Newton converge monotonically (no overshoot)? **A.** $\partial^2 C / \partial \sigma^2 \propto d_1 d_2$, which vanishes at $\hat{\sigma} = \sqrt{2|\ln(S_t/K) + r\tau|/\tau}$: C is convex for $\sigma < \hat{\sigma}$, concave for $\sigma > \hat{\sigma}$, so $\hat{\sigma}$ is the inflection point and where Vega $= C'$ is maximal. Starting there, the contraction ratio $1 - F'(\hat{\sigma})/F'(\hat{\sigma}) \in (0,1)$ because $F'(\hat{\sigma})$ is the maximum slope, so the first step cannot overshoot the root; the iterates then stay on the monotone (concave) side and converge monotonically and quadratically.
6. Put implied vol: why must a put and a call at the same (K,T) have the same implied vol? Derive it from put-call parity. **A.** Parity $C - P = S_t - Ke^{-r\tau}$ is σ -independent, so $P(\sigma) = C(\sigma) - (S_t - Ke^{-r\tau})$. Hence $\partial P / \partial \sigma = \partial C / \partial \sigma = \text{Vega} > 0$: the put has the same vega and strict monotonicity. Since the market call and put at (K,T) satisfy parity, the σ that reprices the call also reprices the put — one implied vol per (K,T) .
7. Describe the implied-vol smile/skew in the strike direction and the term structure in maturity. What market fact does each encode? **A.** Strike direction: a *smile* (convex U, high vol in both tails) encodes excess kurtosis / heavy tails vs lognormal; a *skew/smirk* (decreasing in K , high vol for low-strike puts) encodes negative skewness / crash risk, the typical equity-index shape. Maturity direction (term structure): flat under constant σ ; under time-dependent $\sigma(t)$, $\sigma^{\text{impl}}(T)^2 = (1/T) \int_0^T \sigma(t)^2 dt$ (RMS average), so a rising instantaneous vol gives an upward-sloping term structure. Generic surface: pronounced short-maturity smile flattening to a milder long-maturity skew.

8. Why does constant-vol Black-Scholes fail to fit the whole surface? State precisely what a single σ cannot reproduce. **A.** If BS held, σ^{impl} would be one constant for all (K,T) — a flat plane. The market surface is curved and tilted, so a single σ cannot reproduce simultaneously the strike-direction smile/skew (heavy, asymmetric tails) and the maturity term structure (vol clustering / time-varying variance). Constant volatility and Gaussian returns are the broken assumptions.
9. Distinguish implied vs local vs stochastic volatility: which is a quoting convention, which is a deterministic $\sigma(S,t)$, which adds its own SDE? How does this motivate Heston (Lecture 7)? **A.** Implied vol is a model-inversion *output* — a re-quoting of price in BS units, one number per quote, saying nothing about true dynamics. Local vol $\sigma(S,t)$ is a deterministic *input* surface (Dupire) calibrated to fit the whole surface today. Stochastic vol $\sigma(W^v,t)$ is a *random* input with its own driving SDE (and correlation ρ with spot). Implied describes the data, local fits it deterministically, stochastic explains it — Heston (random mean-reverting variance, L7) generates smile/skew endogenously and is the COS workhorse.

Lecture 05

1. Derive the CRR up/down/probability parameters by matching the mean and variance of the tree log-return to Black-Scholes. Show every step. **A.** One step: $\ln(S/S_0) = \ln d + \ln(u/d) \cdot R$, $R \sim \text{Bernoulli}(p)$. Mean = $p \ln u + (1-p) \ln d$, Var = $p(1-p)(\ln(u/d))^2$. Match to BS log-return over δt : mean = $(r - \frac{1}{2}\sigma^2)\delta t$, var = $\sigma^2\delta t$. So $p \ln u + (1-p) \ln d = (r - \frac{1}{2}\sigma^2)\delta t$ and $\ln(u/d) = \sigma\sqrt{\delta t}/\sqrt{p(1-p)}$. Closing with $p = \frac{1}{2}$ gives $\ln u - \ln d = 2\sigma\sqrt{\delta t}$ and $\ln u + \ln d = 2(r - \frac{1}{2}\sigma^2)\delta t$, hence $u = \exp(\sigma\sqrt{\delta t} + (r - \frac{1}{2}\sigma^2)\delta t)$, $d = \exp(-\sigma\sqrt{\delta t} + (r - \frac{1}{2}\sigma^2)\delta t)$.
2. Why does the Itô-corrected drift $r - \frac{1}{2}\sigma^2$ (not r) appear inside u and d ? What exactly is being matched — the price or the log-price? **A.** The matching is on the *log-price* increment, whose risk-neutral mean is $(r - \frac{1}{2}\sigma^2)\delta t$ by Itô ($d \ln S = (r - \frac{1}{2}\sigma^2)dt + \sigma dW$). So the drift slot in u, d absorbs the Itô-corrected drift $r - \frac{1}{2}\sigma^2$, not r ; matching $\text{Var}(\ln S) = \sigma^2\delta t$ reproduces the diffusion coefficient.
3. The moment equations give two equations in three unknowns (p, u, d). Which free choice does the deck make, and how does that differ from the classical $d = 1/u$ CRR tree? Do both match the first two moments? **A.** Two equations, three unknowns = one free choice. The deck picks $p = \frac{1}{2}$ (symmetric), giving the u, d above. Classical CRR instead imposes $d = 1/u$ and solves for p , giving $u = e^{\{\sigma\sqrt{\delta t}\}}$, $d = e^{-\{\sigma\sqrt{\delta t}\}}$, $p = (e^{\{r\delta t\}} - d)/(u - d)$. Both match the first two moments to $O(\delta t)$; they differ only in the higher-order/free-parameter choice.
4. State and justify the convergence rate of the binomial method. Why is the convergence non-monotone (sign-oscillating in M)? What is the mechanism? **A.** First order, $|\text{error}| \leq K/M$, but non-monotone (sign oscillates in M). Mechanism: the discrete terminal nodes S_n^M straddle the strike K differently as M changes; the position of K relative to the lattice wobbles irregularly, so the payoff-region sampling error oscillates. This is why a raw tree cannot be naively Richardson-extrapolated.
5. Write the FTCS stencil for $u_t = u_{xx}$, the explicit update, and the matrix form $U^{i+1} = F U^i + p^i$. Why do boundary terms appear only in the first and last entries of p^i ? **A.** Forward time, central space: $(U_j^{i+1} - U_j^i)/k = (U_{j+1}^i - 2U_j^i + U_{j-1}^i)/h^2$, so $U_j^{i+1} = \nu U_{j+1}^i + (1 - 2\nu)U_j^i + \nu U_{j-1}^i$ with $\nu = k/h^2$. In matrix form $U^{i+1} = F U^i + p^i$, $F = \text{tridiag}(\nu, 1 - 2\nu, \nu)$, $p^i = (\nu a, 0, \dots, 0, \nu b)^T$. Boundary terms enter only the first and last entries because only the stencils at $j = 1$ and $j = N_x - 1$ reach into the known boundary nodes U_0, U_{N_x} .
6. Carry out the von Neumann analysis for FTCS. Derive $\xi(\beta) = 1 - 4\nu \sin^2(\beta h/2)$ and show the stability condition reduces to $\nu = k/h^2 \leq \frac{1}{2}$. Which mode is worst? **A.** Insert $U_j^i = e^{i\beta j h}$: $\xi = 1 + \nu(e^{i\beta h} - 2 + e^{-i\beta h}) = 1 - 4\nu \sin^2(\beta h/2)$, using $e^{i\theta} - 2 + e^{-i\theta} = -4\sin^2(\theta/2)$. $\xi \leq 1$ automatically; the binding constraint $\xi \geq -1$ gives $4\nu \sin^2(\beta h/2) \leq 2$. The worst mode is the highest frequency $\beta h = \pi$ ($\sin^2 = 1$), giving $\nu \leq \frac{1}{2}$, i.e. $k \leq h^2/2$.
7. Do the same for BTCS: derive $\xi = 1/(1 + 4\nu \sin^2(\beta h/2))$ and explain why BTCS is unconditionally stable. **A.** Backward time evaluates the space operator at the new level: $\xi - 1 = -4\nu \xi \sin^2(\beta h/2)$, so $\xi = 1/(1 + 4\nu \sin^2(\beta h/2)) \in (0, 1]$ for every $\nu > 0$. Since $|\xi| \leq 1$ for all modes and all k , there is no step restriction — BTCS is unconditionally stable. (CN, $\xi = (1 - 2\nu \sin^2)/(1 + 2\nu \sin^2)$, is likewise unconditionally stable.)
8. What is the CFL-type cost of FTCS's explicitness in terms of how k must scale with h ? Why

is this expensive when refining the grid? **A.** Stability forces $k \leq h^2/2$, so the time step scales quadratically with the space step: halving h forces a fourfold cut in k (and so $\sim 4\times$ the time steps). Refining space therefore makes explicit FTCS very expensive — the cost of explicitness. Implicit BTCS/CN avoid this with one tridiagonal solve per step.

9. Give the local truncation errors of FTCS, BTCS and Crank-Nicolson. Why does CN gain an order in time while costing the same per step as BTCS? **A.** FTCS: $O(k)+O(h^2)$; BTCS: $O(k)+O(h^2)$ (time term opposite sign); CN: $O(k^2)+O(h^2)$. CN is centred at the half time level $i+\frac{1}{2}$, which by symmetry kills the leading $O(k)$ term, leaving $O(k^2)$. It is still one tridiagonal solve per step like BTCS, so it gains an order in time at the same per-step cost — hence preferred (caveat: it can oscillate on non-smooth payoffs).
10. State the Lax equivalence theorem and explain its role: consistency + stability = convergence. Why isn't consistency alone enough? **A.** For a consistent linear finite-difference scheme of a well-posed linear IVP, the scheme converges iff it is stable. Consistency (truncation error $\rightarrow 0$) only says the scheme approximates the PDE locally; without stability, errors (especially high-frequency modes, $|\xi| > 1$) are amplified each step and blow up — so the local approximation does not accumulate to a convergent global solution. Stability bounds the error growth, and the two together give convergence.
11. Transform the reverse-time BS PDE to a constant-coefficient equation via $X = \ln S$, then remove the reaction term via $V = e^{\{-r\tau\}}W$. Show the coefficient algebra ($S^2 \partial_{SS} = \partial_{XX} - \partial_X$). **A.** $X = \ln S$ gives $\partial_S = (1/S) \partial_X$ and $\partial_{SS} = (1/S^2)(\partial_{XX} - \partial_X)$, so $S^2 \partial_{SS} = \partial_{XX} - \partial_X$ and $S \partial_S = \partial_X$. The reverse-time PDE $V_\tau - \frac{1}{2}\sigma^2 S^2 V_{SS} - rSV_S + rV = 0$ becomes $V_\tau - \frac{1}{2}\sigma^2 V_{XX} - (r - \frac{1}{2}\sigma^2) V_X + rV = 0$ (constant coefficients). Then $V = e^{\{-r\tau\}}W$: $V_\tau = e^{\{-r\tau\}}(W_\tau - rW)$, and the rV and $-rW$ terms cancel, leaving $W_\tau - \frac{1}{2}\sigma^2 W_{XX} - (r - \frac{1}{2}\sigma^2) W_X = 0$, a pure advection-diffusion equation.
12. Show that applying BTCS to the log-transformed advection-diffusion equation with the coupling $h^2 = \sigma^2 k$ makes the centre node drop out, yielding $V_{i,j}^{i+1} = \{rk\}(p^* V_{i+1}^{j+1} + (1-p^*) V_{i+1}^{j-1})$. Derive p . **A.** BTCS on $W_\tau = \frac{1}{2}\sigma^2 W_{XX} + (r - \frac{1}{2}\sigma^2) W_X$ produces the centre coefficient $(1/k) W_{j+1}^{i+1} - (\sigma^2/h^2) W_j^{i+1}$. With the coupling
13. In what precise sense is the binomial method “an explicit FD scheme customised to a single time-zero value”? What does a general FD scheme give that the tree throws away? **A.** The recursion above is exactly the risk-neutral binomial backward step (up-prob p , *log-spacing* $\pm \sigma \sqrt{k}$), i.e. an explicit FD scheme on the transformed heat equation, “customised” via $h^2 = \sigma^2 k$ so the centre node drops out and only the two branches remain. The tree returns only V_0^0 (one spot). A general FD scheme returns the value at every* grid node $\{j, i, k\}$, hence prices for all initial spots at once — that whole-grid information is what the tree discards.
14. On the raw S -grid the FTCS diffusion weight is $\frac{1}{2}k\sigma^2 j^2$. What does this do to stability near $S=L$, and how does working in $x = \ln S$ fix it? Connect this to domain-truncation pitfalls (cf. the COS $[a, b]$ range at small c_2). **A.** The weight $\frac{1}{2}k\sigma^2 j^2$ grows like j^2 , so the effective stability condition $\sigma^2 j^2 k < \sim 1$ is hardest at the largest j (S near the truncation L): stability degrades quadratically near the upper boundary. Working in $x = \ln S$ makes the PDE constant-coefficient, so the stencil weight is uniform and this blow-up disappears. Both are the same lesson as the COS $[a, b]$ trap: a method on a finite window is only as good as the window — FD truncates the state space, COS truncates the Fourier-cosine range, and at small $c_2 = \sigma^2 T$ that COS window collapses just as the FD domain must be chosen to hold all the mass.
15. The deck's Slide 57 prints a garbled p^* formula. Why is it wrong, and what is the correct dimensionless form $p = \frac{1}{2}(1 + (r - \sigma^2/2)\sqrt{k}/\sigma)$? Check the $k \rightarrow 0$ limit. **A.** The slide prints $\frac{1}{2}(1 + \sqrt{k}/\sigma \cdot r - \sigma^2)$, which is dimensionally inconsistent (the $-\sigma^2$ is not scaled and a factor is missing). The correct algebra gives $p = \frac{1}{2}(1 + (r - \frac{1}{2}\sigma^2)\sqrt{k}/\sigma)$, which is dimensionless (the $\sqrt{k}/\sigma$ carries units that cancel). As $k \rightarrow 0$ the correction $\rightarrow 0$ and $p^* \rightarrow \frac{1}{2}$, consistent with the symmetric $p = \frac{1}{2}$ choice of the moment-matched tree. Treat the slide as a transcription error.

Cluster 2 — Monte Carlo

Lecture 04

1. Construct the MC integral estimator and prove it is unbiased. Derive $\text{Var}(\bar{X}_N) = \sigma^2/N$ from independence and state the standard error. **A.** Write $I = \int h(x) g(x) dx = E_g[h(X)]$, estimate by $\bar{X}_N = (1/N) \sum h(x_i)$, $x_i \sim g$ i.i.d. Unbiased: $E[\bar{X}_N] = (1/N) \sum E[h(X)] = I$. Variance:

- $\text{Var}(\bar{X}_N) = (1/N^2) \Sigma \text{Var}(X_i) = (1/N^2) \cdot N \sigma^2 = \sigma^2/N$ (cross-covariances vanish by independence). Standard error $\text{se} = \sigma/\sqrt{N} = O(N^{-1/2})$.
- Why is the MC error $O(N^{-1/2})$ independent of dimension, while tensor-product quadrature is $O(N^{-k/d})$? When does MC win? **A.** The CLT rate $\text{se} = \sigma/\sqrt{N}$ depends only on the number of samples N and the variance σ^2 , not on the dimension d — sampling a high-dimensional density is no harder per draw. Tensor-product quadrature of order k uses $N = m^d$ points (m per axis) for $O(m^{-k}) = O(N^{-k/d})$, which degrades as d grows (curse of dimensionality). MC wins in high dimension: baskets, many time steps, path-dependent payoffs.
 - Write a CLT-based confidence interval for the MC estimate. Why use the sample variance s_N^2 with $(N-1)$, and what does it cost? **A.** By the CLT $(\bar{X}_N - I)/(\sigma/\sqrt{N}) \rightarrow N(0,1)$, so a $(1-\alpha)$ CI is $\bar{X}_N \pm z\{1-\alpha/2\} \cdot \sigma/\sqrt{N}$ ($z=1.96$ for 95%). σ is unknown, so use $s_N^2 = (1/(N-1)) \Sigma (X_i - \bar{X}_N)^2$; the $N-1$ (Bessel) makes it an *unbiased* variance estimate, at the cost of one degree of freedom (negligible for large N). The reported half-width $z \cdot s_N/\sqrt{N}$ is exactly what variance reduction tries to shrink.
 - State the Koksma-Hlawka inequality (star discrepancy + Hardy-Krause variation). Why can QMC achieve near- $O(N^{-1}) = (\log N)^d/N$? **A.** $|(1/N) \Sigma f(x_i) - \int f| \leq V(f) \cdot D_N^*$, where $V(f)$ is the Hardy-Krause variation (integrand roughness, independent of points) and D_N^* is the star discrepancy (worst-case box gap, independent of f). A low-discrepancy sequence (Sobol, Halton) has $D_N^* = O((\log N)^d/N)$, so for bounded-variation f the deterministic QMC error is $O((\log N)^d/N)$ — essentially $O(N^{-1})$, a full power faster than plain MC's $O(N^{-1/2})$.
 - What is “effective dimension” and why can plain QMC degrade in high nominal dimension? What does randomised QMC recover? **A.** The $(\log N)^d$ factor is enormous for large d , so the QMC advantage only shows at impractical N unless the *effective* dimension (how much each coordinate actually matters) is low — Brownian-bridge / PCA path constructions lower it and make QMC work for option pricing. Plain QMC is deterministic, giving no error estimate; randomised QMC (e.g. random shifts of a lattice rule) restores an unbiased estimator and lets you estimate the error from independent shifts while keeping most of the discrepancy gain.
 - Write the Euler-Maruyama scheme for $dX = a dt + b dW$. What iterated Itô integral does it drop? **A.** $\hat{X}_{t_k} = \hat{X}_{t_{k-1}} + a(t_{k-1}, \hat{X})h + b(t_{k-1}, \hat{X})\sqrt{h} Z_k$, $Z_k \sim N(0,1)$, freezing coefficients at the left endpoint (Itô convention). It drops the iterated Itô integral $\int \int dW dW = \frac{1}{2}[(\Delta W)^2 - h]$, i.e. the variation of b *within* the step — the leading correction that Milstein restores.
 - Derive the Milstein correction from the Itô-Taylor expansion: show $\int \int dW dW = \frac{1}{2}[(\Delta W)^2 - h]$ and the resulting $\frac{1}{2}bb'[(\Delta W)^2 - h]$ term. **A.** Expand $\sigma(X_u)$ by Itô around X_s : $d\sigma(X_t) = \dots dt + \sigma' \sigma dW$, so $\sigma(X_u) = \sigma(x) + \sigma'(x)\sigma(x) \int dW + O(dt)$. Substituting into $\int \sigma dW$ gives the extra term $\sigma' \sigma \cdot \int \sigma^u \{t_k\} \int \sigma^u dW_r dW_u$. The inner integral is $\int (W_u - W_s) dW_u = \frac{1}{2}[(\Delta W)^2 - h]$. So the update gains $\frac{1}{2}\sigma\sigma'[(\Delta W)^2 - h] = \frac{1}{2}bb'(Z^2 - 1)h$. It vanishes for additive noise (b constant), where Euler already has strong order 1.
 - Define strong vs weak convergence order. Give the orders for Euler $(\frac{1}{2}, 1)$ and Milstein $(1, 1)$. Why does Milstein buy pathwise but not distributional accuracy? **A.** Strong order β : $E|\hat{X}_{t_K} - X_T| \leq ch^\beta$ (pathwise, same Brownian path). Weak order β : $|E[f(\hat{X}_{t_K})] - E[f(X_T)]| \leq ch^\beta$ for smooth f (laws only). Euler: strong $\frac{1}{2}$, weak 1; Milstein: strong 1, weak 1. Milstein improves the *strong* (pathwise) rate but has the same weak rate as Euler — so for a European price $E[f(X_T)]$ it buys nothing distributionally; it matters for path-dependent payoffs and for coupling coarse/fine levels (multilevel MC) where pathwise accuracy is used.
 - Antithetic variates: derive the variance of the antithetic estimator and state the monotonicity/comonotonicity condition for it to help. **A.** Pair each draw U with its reflection $1-U$: $\hat{I} = (1/M) \Sigma \frac{1}{2}(f(U_i) + f(1-U_i))$, unbiased. Per pair, $\text{Var}(\frac{1}{2}(f(U) + f(1-U))) = \frac{1}{2}\text{Var}(f(U)) + \frac{1}{2}\text{Cov}(f(U), f(1-U))$. Comparing with the plain estimator on the same $2M$ evaluations, antithetic wins iff $\text{Cov}(f(U), f(1-U)) < 0$, which is guaranteed when f is monotone (by the FKG/correlation lemma: f increasing $\Rightarrow -f(1-\cdot)$ increasing $\Rightarrow \text{Cov}(f(U), f(1-U)) \leq 0$). It can backfire for strongly non-monotone payoffs (e.g. a straddle).
 - Control variates: derive the optimal coefficient $\theta = \text{Cov}(X, Y)/\text{Var}(Y)$ and show the variance is reduced by $(1-\rho^2)$. What is a good control for an Asian option? **A.** With $Z_\theta = X + \theta(E[Y] - Y)$, $E[Z_\theta] = E[X]$ for all θ (unbiased), $\text{Var}(Z_\theta) = \text{Var}(X) - 2\theta\text{Cov}(X, Y) + \theta^2\text{Var}(Y)$. Minimising the quadratic gives $\theta = \text{Cov}(X, Y)/\text{Var}(Y)$, and substituting, $\text{Var}(Z_\theta) = (1-\rho^2_{XY})\text{Var}(X)$. The

reduction is dramatic when $|\rho|$ is near 1. For an arithmetic-average Asian (no closed form), use the geometric*-average Asian (closed form for equally spaced fixings) as control — the two payoffs are nearly perfectly correlated path by path (Kemna-Vorst).

11. Importance sampling: write the likelihood-ratio estimator and the Gaussian drift-shift. What is the infinite-variance pitfall, and how does Girsanov connect? **A.** Sample from a proposal q and reweight: $E_p[f(X)] = E_q[f(X)p(X)/q(X)]$, estimator $(1/N)\sum f(X_i)p(X_i)/q(X_i)$, $X_i \sim q$, with p/q the likelihood ratio (dp/dq) . Gaussian shift $X=Z+\theta$: $V=E[f(Z+\theta)\exp(-\theta^T Z - \frac{1}{2}\|\theta\|^2)]$, moving mass into the payoff region (e.g. so $S_T > K$ is typical for a deep-OTM call). Pitfall: a proposal with too-light tails makes p/q blow up and the estimator has *infinite variance* — worse than plain MC; IS is the only one of the four that can backfire catastrophically. In continuous time the reweighting is exactly Girsanov: shifting the drift of W by Y_s corresponds to the density process $\exp(\int Y_s dW - \frac{1}{2}\int Y_s^2 ds)$.
12. Why is MC the benchmark that COS and FFT are compared against, and on what axes (rate, dimension, smoothness) does COS dominate for European options? **A.** MC is unbiased by construction and dimension-robust, so it is the brute-force reference: a fast method is “right” if it agrees with the MC price to within the CLT standard error. For a 1-D European under a model with known smooth characteristic function, COS converges *exponentially* in the number of cosine terms, vs MC’s $O(N^{\{-1/2\}})$ and FFT’s algebraic quadrature rate — reaching machine precision with a few hundred terms where MC needs $\sim 10^{\{12\}}$ paths. MC still wins on high-dimensional baskets and path-dependent payoffs with no transform structure.

Cluster 3 — Fourier / COS

COS-specific drilling lives in `cos-deep-dive.md`. This section holds the surrounding Fourier-family theory from Lecture 6 that Fang can ask “around” COS.

Lecture 6 — the Fourier family (around COS)

1. State the continuous Fourier transform and its inverse. Which scaling/sign convention did the slides use, and why does the convention not matter for the method? **A.** One convention: $\hat{f}(\omega) = \int f(x)e^{i2\pi\omega x} dx$, $f(x) = (1/2\pi) \int \hat{f}(\omega)e^{-i2\pi\omega x} d\omega$. The slides place the 2π and sign so that the characteristic function φ_X is (up to that 2π -scaling) the Fourier transform of the density. The convention does not matter for COS because only internal consistency is needed — COS reads cosine coefficients off φ at the grid frequencies $k\pi/(b-a)$, and any consistent choice gives the same A_k .
2. Define the characteristic function $\varphi_X(t)$. In what precise sense is it the “Fourier dual” of the density f_X ? Write the inversion formula $f_X(x) = (1/2\pi) \int e^{-itx} \varphi_X(t) dt$. **A.** $\varphi_X(t) = E[e^{itX}] = \int e^{itx} f_X(x) dx$ — this is the Fourier transform of the density (up to the 2π convention), so f_X and φ_X are a Fourier pair. The inversion is $f_X(x) = (1/2\pi) \int e^{-itx} \varphi_X(t) dt$. This is exactly the integral COS evaluates cheaply, without quadrature.
3. Why are characteristic-function methods attractive at all? Name four models whose ch.f. is known in closed form but whose density is not. **A.** Because for a huge class of models the density f is intractable but the characteristic function φ is known in closed (or semi-closed) form, and a method needing only φ inherits that whole library for free. Examples: Heston, Variance-Gamma, Merton/Kou jump-diffusions (and the affine jump-diffusions generally), CGMY/NIG Lévy models.
4. Write the DFT and inverse DFT. What does the FFT change about the cost, from what to what, and by what factorisation idea? Does COS use the FFT? **A.** $DFT X_k = \sum_n x_n e^{i2\pi kn/N}$, inverse $x_n = (1/N) \sum_k X_k e^{-i2\pi kn/N}$. Naively $O(N^2)$; the FFT factorises the DFT matrix into $O(\log N)$ sparse factors (radix-2 split into even/odd sub-DFTs of length $N/2$), giving $O(N \log N)$. European COS does *not* use the FFT — its cost is the bare $O(N)$ evaluation of one cosine sum (the FFT returns only for the discrete-barrier/Bermudan recursions).
5. Derive the half-range cosine series from the full Fourier series via the even extension. Why does the even extension kill the sine terms? **A.** A function on $[0, L]$ has a full Fourier series with sin and cos terms. Extending it *evenly* to $[-L, L]$ makes it an even function; since sin is odd,

- all sine coefficients $b_n = (2/L) \int_0^L f(x) \sin(n\pi x/L) dx$ vanish, leaving the half-range cosine series $f(x) = a_0/2 + \sum_{n=1}^{\infty} a_n \cos(n\pi x/L)$, $a_n = (2/L) \int_0^L f(x) \cos(n\pi x/L) dx$. COS uses this on $[a, b]$ (shift $x \mapsto x-a$, period $2(b-a)$); the cosine basis gives real coefficients read directly off φ .
6. Carr-Madan: why is the undamped call $C_T(k)$ not square-integrable in log-strike k ? What is the role of the damping factor $e^{\{\alpha k\}}$, and what is the cost of having a free parameter α ? **A.** As $k = \ln K \rightarrow -\infty$, $C_T(k) \rightarrow S_0$ (a nonzero constant), so C_T is not in L^1/L^2 and cannot be Fourier-transformed directly. Damping $c_T(k) = e^{\{\alpha k\}} C_T(k)$ with $\alpha > 0$ makes it integrable so its Fourier transform exists. The cost is the free parameter α : it has no canonical value, its choice affects accuracy/stability, and it must be tuned. COS needs no such parameter (the range comes from cumulants).
 7. Carr-Madan: $\psi_T(v) = e^{\{-rT\}} \varphi_X(v - (\alpha+1)i) / (\alpha^2 + \alpha - v^2 + i(2\alpha+1)v)$. Why is it a selling point that ψ_T is expressible through the ch.f.? How is the price recovered, and by what numerical tool? **A.** Because the transform of the damped call ψ_T is written purely through φ_X (evaluated at the shifted complex argument $v - (\alpha+1)i$), the whole library of models with known φ is immediately available — no density needed. The price is recovered by the inverse transform $C_T(k) = (e^{\{-\alpha k\}/(2\pi)}) \int_{-\infty}^{\infty} e^{-ivk} \psi_T(v) dv$, evaluated numerically by the FFT (a trapezoidal-type quadrature of an oscillatory integrand). The two weaknesses vs COS: it still does a numerical integration, and it needs α .
 8. Contrast COS and Carr-Madan on: core operation, free parameters, convergence rate for smooth f , and number of terms to reach machine precision. **A.** COS: one $O(N)$ cosine sum, no integration; no free parameter (range from cumulants); exponential convergence for smooth f ; dozens-hundreds of terms to machine precision; analytic Greeks from the same sum. Carr-Madan/FFT: an FFT quadrature $O(N \log N)$; free damping α to tune; algebraic (quadrature-limited) convergence; thousands of nodes for far worse accuracy. Both need only φ .
 9. Give the BS log-price characteristic function and state the risk-neutral drift $\mu = r - \frac{1}{2}\sigma^2$. Where does the martingale condition fix μ ? **A.** $\varphi(u) = \exp(iu\mu T - \frac{1}{2}\sigma^2 u^2 T)$ with $\mu = r - \frac{1}{2}\sigma^2$ for $X_T = \ln(S_T/S_0)$. The drift is fixed by the martingale condition that $e^{\{-rt\}} S_t$ be a Q-martingale: with $S_T = S_0 \exp(\mu T + \sigma W_T)$, $E[e^{\{-rT\}} S_T] = S_0$ requires $e^{\{\mu T\}} E[e^{\{\sigma W_T\}}] = e^{\{rT\}}$, i.e. $\mu + \frac{1}{2}\sigma^2 = r$, so $\mu = r - \frac{1}{2}\sigma^2$. (This is the same $-\frac{1}{2}\sigma^2$ Itô correction as in $d \ln S$.)

Lecture 10 — barrier options (theory around COS)

Deep COS-for-barrier drilling is in `cos-deep-dive.md` (sets H-K). These are the surrounding theory items.

10. Name the 8 barrier types and explain why exactly 8. State in-out parity and its uses. **A.** up/down $\times$ in/out $\times$ call/put = $2 \times 2 \times 2 = 8$: up-and-out call, up-and-in call, down-and-out call, down-and-in call, and the four put analogues. In-out parity: knock-in + knock-out (same K, B, T) = vanilla, because exactly one of the two is alive at maturity — $C_{in}(t) + C_{out}(t) = e^{\{-r(T-t)\}} E[(S_T - K)^+]$. Uses: price the easier one (usually the knock-out) plus the vanilla and back out the other by subtraction; and a free correctness check on any numerical barrier price.
11. Write the barrier price as (i) a first-passage expectation and (ii) a localized Feynman-Kac PDE. How does the barrier enter the PDE? **A.** (i) $V(x, t) = e^{\{-r(T-t)\}} E_x[(\alpha(e^{X_T} - K))^+ + 1_{\{\tau_B > T\}}]$ — discounted vanilla payoff times a survival indicator. (ii) On $[a, b]$, V solves the localized BS PDE $V_t + (r - \frac{1}{2}\sigma^2)V_x + \frac{1}{2}\sigma^2 V_{xx} - rV = 0$ with terminal data the vanilla payoff and *absorbing (Dirichlet) boundary* $V=0$ at the knock-out level(s). The barrier enters not through the equation but through the homogeneous boundary condition: hitting the barrier sets value to zero.
12. State and prove the reflection principle result $P(\tau_a \leq T) = 2P(W_T \geq a)$. Hence write the density of the running maximum. **A.** By the strong Markov property at τ_a and BM symmetry, a path that has touched a is equally likely to end above or below a : $P(W_T \geq a | \tau_a \leq T) = \frac{1}{2}$. So $P(\tau_a \leq T) = P(\tau_a \leq T, W_T \geq a) + P(\tau_a \leq T, W_T < a) = 2P(\tau_a \leq T, W_T \geq a) = 2P(W_T \geq a)$ (since reaching $W_T \geq a$ forces $\tau_a \leq T$). Hence the running max $\Xi = \max W$ has the law of $|W_T|$, with density $p_{\Xi}(a) = \sqrt{(2/\pi T)} \cdot e^{\{-a^2/2T\}}$, $a \geq 0$.
13. Derive the joint density of (running max, W_T). Why does the reflection at level $2a-b$

- appear? **A.** Reflect the path after τ_a across level a : for $b \leq a$, $P(\bar{E} \geq a, W_T < b) = P(\bar{E} \geq a, W_T > 2a - b) = P(W_T > 2a - b)$ (since $2a - b \geq a$ forces the max to exceed a). Differentiating the tail $P(\bar{E} > a, W_T \leq b) = P(W_T \geq 2a - b)$ twice gives $p_{\{\bar{E}, W_T\}}(a, b) = \sqrt{(2/\pi)} \cdot ((2a - b)/T)^{3/2} \cdot e^{-((2a - b)^2/2T)}$, $a \geq \max(b, 0)$. The $2a - b$ is the mirror image of the endpoint b reflected across the barrier a — the reflected path's terminal point. (Drift is handled by a Girsanov reweighting.)
14. Method of images: state $U(S, t) = S^{\alpha} V(B^2/S, t)$, $\alpha = \frac{1}{2}(1 - 2r/\sigma^2)$, and the down-and-out call it gives. **A.** One checks directly that if $LV = 0$ then the image $U(S, t) = S^{\alpha} V(B^2/S, t)$, $\alpha = \frac{1}{2}(1 - 2r/\sigma^2)$, also solves $LV = 0$ — the BS analogue of reflecting a heat-equation solution across a wall. Subtracting it with the right weight cancels the value on $S = B$, giving the down-and-out call $C_{DO}(S_t, K) = C_{van}(S_t, K) - (S_t/B)^{\alpha} C_{van}(B^2/S_t, K)$: the vanilla minus the image evaluated at the mirror spot B^2/S_t with weight $(S_t/B)^{\alpha}$, which removes exactly the barrier-crossing paths.
15. Sine-series PDE route: what substitution turns the localized BS PDE into the heat equation, and why the sine (not cosine) basis? **A.** With $\tau = \frac{1}{2}\sigma^2(T - t)$, $\kappa = 2r/\sigma^2$, $V = e^{\alpha x + \beta \tau} U$, $\alpha = -\frac{1}{2}(\kappa - 1)$, $\beta = -\alpha^2 - \kappa$, $z = x - a$, the localized BS PDE becomes the heat equation $U_\tau = U_{zz}$ on $[0, b - a]$ with $U(0, \tau) = U(b - a, \tau) = 0$. The homogeneous *Dirichlet* (zero) boundary conditions select the *sine* basis ($\sin(n\pi z/(b - a))$ vanishes at both ends, $\cos$ does not); each mode propagates by $e^{-((n\pi/(b - a))^2 \tau)}$. This is the structural precursor of COS — but the propagator is the explicit heat semigroup, whereas COS uses φ and so generalises to any model.
16. Why is a discretely-monitored barrier a backward recursion? Where does the Markov property enter, and why does the FFT reappear (vs European COS)? **A.** Survival is a product of indicators $\prod 1_{\{x_m < h\}}$, and by the Markov property the joint density factorises into one-step transitions $f(x_M | x_{M-1}) \dots f(x_1 | x_0)$, so the M -dimensional integral *nests* into a backward recursion: at each date $c(x, t_{m-1}) = e^{-r\Delta t} \int v(y, t_m) f(y | x) dy$ (European-COS form), then knock out by overwriting $v = R$ above h , which splits the coefficient V_k at h . The split feeds the previous V_j through an analytic coupling $M_{\{k, j\}}$ that is Hankel-plus-Toeplitz, i.e. a discrete convolution — so each step is done by FFT in $O(N \log N)$, overall $O((M - 1)N \log_2 N)$. European COS needs no FFT (single $O(N)$ sum); the barrier needs the whole coefficient vector propagated every step, which is the convolution the FFT accelerates.

Cluster 4 — Models (Heston / jumps)

Lecture 07

- Write down the Heston SDE system under Q , naming every parameter, and state what $dW^S dW^v = \rho dt$ means physically. **A.** $dS_t = rS_t dt + \sqrt{v_t} S_t dW^S_t$, $dv_t = \kappa(\bar{v} - v_t)dt + \gamma\sqrt{v_t} dW^v_t$, with $dW^S dW^v = \rho dt$. Here v_t is the instantaneous *variance* (vol is $\sqrt{v_t}$), $\bar{v}$ the long-run variance, $\kappa > 0$ the mean-reversion speed, $\gamma > 0$ the vol-of-vol (smile curvature), ρ the spot/variance correlation (skew slope). $dW^S dW^v = \rho dt$ means shocks to price and to variance are instantaneously correlated; $\rho < 0$ (the leverage effect) means down-moves in S come with up-moves in variance, steepening the skew.
- Derive the log-spot form $d \ln S_t$ and show how the Cholesky factor decorrelates the two Brownian motions into independent W^S, W^v . **A.** Itô on $\ln S$ with $(dS/S)^2 = v dt$: $d \ln S_t = (r - \frac{1}{2}v_t)dt + \sqrt{v_t} dW^S_t$. For $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} = LL^T$, $L = \begin{bmatrix} 1 & 0 \\ \rho & \sqrt{1 - \rho^2} \end{bmatrix}$, write the correlated pair via independent $(\tilde{W}_1, \tilde{W}_2)$: $W^S = \tilde{W}_1$, $W^v = \rho \tilde{W}_1 + \sqrt{1 - \rho^2} \tilde{W}_2$, which checks $\text{Cov}(W^{S, v}) = \rho t$. Equivalently, driving the variance by its own clean W^v , the log-spot diffusion becomes $\sqrt{v_t}(\rho dW^{v, v} + \sqrt{1 - \rho^2} dW^\perp)$ with $W^{\perp, W} \perp$ — the form used for Euler and the affine derivation.
- State the conditional law of ν_T given ν_t (noncentral chi-squared): give d , the noncentrality λ , and the scaling c . **A.** $\nu_T | \nu_t \sim Y/(2c)$ with $Y \sim \chi^2_{d(\lambda)}$ noncentral chi-squared, $c = 2\kappa/(\gamma^2(1 - e^{-\kappa\tau}))$, $\tau = T - t$, degrees of freedom $d = 4\kappa\bar{v}/\gamma^2$, noncentrality $\lambda = 2c e^{-\kappa\tau} \nu_t$. So the variance is exactly a scaled noncentral χ^2 — this is what the Broadie-Kaya exact scheme samples.
- State the Feller condition precisely. What exactly does it guarantee, and what happens to ν_t when it is violated? **A.** Feller: $2\kappa\bar{v} \geq \gamma^2$. It guarantees the origin is inaccessible — ν_t stays strictly positive a.s. If violated, ν_t can reach 0 (and reflects/leaves again for the pure CIR

process), but it never goes negative. Practically the exact noncentral- χ^2 scheme is guaranteed well-defined ($d > 0$) under Feller.

5. Derive the Feller condition from the noncentral chi-squared density blowing up at zero (use $I_{\nu}(y) \sim (y/2)^{\nu} / \Gamma(\nu+1)$ as $x \rightarrow 0$). **A.** The noncentral χ^2 density $\propto x^{\{d/4-1/2\}} I_{\{d/2-1\}}(\sqrt{\lambda x})$. As $x \rightarrow 0$, $I_{\{d/2-1\}}(\sqrt{\lambda x}) \sim (\sqrt{\lambda x}/2)^{\{d/2-1\}} / \Gamma(d/2)$, so the density behaves like $x^{\{d/4-1/2\}} \{(\sqrt{\lambda x}/2)^{\{d/2-1\}}\} = x^{\{d/2-1\}}$. For it not to blow up at the origin we need $d/2-1 \geq 0$, i.e. $d = 4\kappa\bar{\nu}/\gamma^2 \geq 2$, which is $2\kappa\bar{\nu} \geq \gamma^2$ — the Feller condition. A large γ relative to the restoring force $\kappa\bar{\nu}$ makes excursions to zero likely.
6. Is the Heston model affine in (S, ν) ? Show why not, then show how moving to $\ln S$ restores the affine condition (give K_0, K_1, H_0, H_1). **A.** Not affine in (S, ν) : the diffusion covariance has top-left entry νS^2 , not affine in the state. Switching to $x = \ln S$, the drift becomes $K_0 = (r, \kappa\bar{\nu})^T + K_1 \cdot (x, \nu)^T$ with $K_1 = [[0, -1/2], [0, -\kappa]]$ (affine), and $\sigma\sigma^T = \nu \cdot [[1, \gamma\rho], [\gamma\rho, \gamma^2]]$, i.e. $H_0 = 0$ and H_1 the tensor whose ν -slice is $[[1, \gamma\rho], [\gamma\rho, \gamma^2]]$ and x -slice is 0 — affine in ν . So in log-coordinates Heston is affine and the Duffie-Pan-Singleton transform applies.
7. Set up the affine ansatz $\phi = \exp(\alpha + \beta \cdot x)$ and derive the Riccati ODE system. Where does each ODE come from when you substitute into the Feynman-Kac PDE? **A.** Feynman-Kac: ϕ solves $\partial_t \phi + L\phi - R\phi = 0$, $\phi(T, x) = e^{\{u \cdot x\}}$. Substituting $\phi = e^{\{\alpha + \beta \cdot x\}}$ gives $\partial_{\{x_i\}} \phi = \beta_i \phi$, $\partial_{\{x_i x_j\}} \phi = \beta_i \beta_j \phi$, $\partial_t \phi = (\dot{\alpha} + \beta \cdot \dot{x}) \phi$. Dividing by ϕ and using that $\mu, \sigma\sigma^T, R$ are affine in x , collect terms: the part *linear* in x must vanish, giving $\beta = \rho_1 - K_1^T \beta - 1/2 \beta^T H_1 \beta$ (the Riccati ODE); the part *constant* in x gives $\dot{\alpha} = \rho_0 - K_0 \cdot \beta - 1/2 \beta^T H_0 \beta$. Boundary conditions $\beta(T, T) = u$, $\alpha(T, T) = 0$. Affineness is what lets the x -dependence separate cleanly.
8. Solve the scalar CIR Riccati $\dot{\beta} = -\kappa \beta + (1/2) \gamma^2 \beta^2$ explicitly via $g = 1/\beta$. Then integrate for α . **A.** Set $g = 1/\beta$; then $\dot{g} = -\beta/\beta^2 = \kappa g - 1/2 \gamma^2$, a linear ODE. With integrating factor $e^{\{-\kappa t\}}$: $d(g e^{\{-\kappa t\}}) = -1/2 \gamma^2 e^{\{-\kappa t\}} dt$, so $g(t) = e^{\{\kappa t\}} (g(0) - \gamma^2 (1 - e^{\{-\kappa t\}}) / (2\kappa))$, $g(0) = 1/u$. Inverting, $\beta(t) = u e^{\{-\kappa t\}} / (1 - 1/2 u \gamma^2 \xi_{\kappa}(t))$ with $\xi_{\kappa}(t) = (1 - e^{\{-\kappa t\}}) / \kappa$. Then $\dot{\alpha} = \kappa \bar{\nu} \beta$ integrates (since $e^{\{-\kappa s\}} = \xi_{\kappa}'(s)$) to $\alpha(t) = -(2\kappa \bar{\nu} / \gamma^2) \ln(1 - 1/2 u \gamma^2 \xi_{\kappa}(t))$. This is the same Riccati mechanism behind the full Heston cf.
9. Write the closed-form Heston cf of $\ln S_T$: give b, a, s and the formulas for $\beta(\tau, u)$ and $\alpha(\tau, u)$. **A.** With $b = \gamma \rho u - \kappa$, $a = u(1 - u)$, $s = \sqrt{(b^2 + a \gamma^2)}$: $\beta(\tau, u) = a(1 - e^{\{-s\tau\}}) / (2s - (s+b)(1 - e^{\{-s\tau\}}))$, $\alpha(\tau, u) = -r\tau + ru\tau - \kappa \bar{\nu} [(s+b)\tau/\gamma^2 + (2/\gamma^2) \ln(1 - ((s+b)/2s)(1 - e^{\{-s\tau\}}))]$. Then $\phi_{\{\ln S_T\}}(\omega; \tau) = \exp(\alpha(\tau, i\omega) + \ln S_t + \beta(\tau, i\omega) \nu_t)$. This is the load-bearing object COS consumes.
10. Explain the “little Heston trap”: what goes wrong at long maturities, what causes it (branch cut of complex log / sqrt), and which grouping fixes it? **A.** The cf involves a complex $\sqrt{s}$ and complex log, which are multi-valued. The original Heston (1993) grouping (with $g = (b+s)/(b-s)$ and $e^{\{+s\tau\}}$) repeatedly crosses the principal branch cut of $\ln$ as τ grows, making the integrand discontinuous and giving wildly wrong long-maturity prices. The algebraically equivalent “little trap” grouping (Albrecher et al. 2007) — written with $e^{\{-s\tau\}}$ and $1 - ((s+b)/2s)(1 - e^{\{-s\tau\}})$, the form above, taking the principal $\sqrt{\cdot}$ with non-negative real part — stays on the principal branch and is stable for all τ .
11. How does the Heston cf enter the COS pricing formula? At which frequencies is it evaluated, and why is Heston the ideal use case for COS? **A.** The Heston ϕ is the *only* model-specific input to the COS formula $v \approx e^{\{-r\tau\}} \sum' \text{Re}\{\phi(k\pi/(b-a)) e^{\{-ik\pi a/(b-a)\}}\} V_k$, evaluated at the COS frequencies $\omega_k = k\pi/(b-a)$ (the V_k are the model-free payoff coefficients). Heston is the ideal use case because its *density* is unknown in closed form but its cf is known closed-form — exactly the regime where COS (which never touches the density) dominates, reaching $\sim 10^{-10}$ error with $N \approx 160$ terms.
12. Under Heston, how do you choose the COS truncation range $[a, b]$? Why is the short-maturity (small c_2) case dangerous, and how does negative ρ make it worse? **A.** Same cumulant rule $[a, b] = c_1 \pm L \sqrt{(c_2 + \sqrt{c_4})}$, with c_1, c_2, c_4 of $\ln S_T$ obtained by differentiating $\log \phi$ at $\omega = 0$. Danger at short τ : c_2 (the variance) is small, so the box collapses toward c_1 ; if the higher cumulants are under-weighted the range chops off mass and COS loses spectral accuracy — adding terms N does not help (it is a range, not a series, error). Negative ρ produces a heavy left tail (skew) plus γ fattens the tails, so the true support is wider than a c_2 -only box; the fix is a larger L (10–12) and keeping the $\sqrt{c_4}$ term, not the BS-style $L = 8$ –10 with c_4 ignored. This is the Assignment-2 bug generalised.
13. Why can the Euler scheme produce negative variance? List truncation, reflection, and log-

- variance fixes and state the drawback of each. **A.** Euler updates $v_{i+1} = v_i + \kappa(\bar{v} - v_i)\Delta + \gamma\sqrt{v_i}/\sqrt{\Delta} Z$ with Z Gaussian and unbounded below, so $P(v_{i+1} < 0) > 0$ always; then $\sqrt{v}$ is undefined. Fixes: (i) truncation/floor $v \leftarrow \max(v, \varepsilon)$ — simple but introduces truncation bias; (ii) reflection $v \leftarrow |v|$ — useful when Feller fails but the reflected path sits systematically above the floored one near zero; (iii) simulate $\ln v$ by Euler and exponentiate — does *not* work, $\ln v$ can blow up so $v = e^{\{\ln v\}}$ explodes. All are biased, worst when Feller is badly violated.
14. State the Broadie-Kaya exact simulation algorithm (4 steps). Which step is hard and how is it solved (cf of integrated variance $\rightarrow$ COS inversion $\rightarrow$ inverse sampling)? **A.** (1) Sample $v_t | v_u$ from the noncentral χ^2 . (2) Sample $\int_u^t v_s ds$ given (v_u, v_t) — the hard step. (3) Recover $\int \sqrt{v} dW \wedge v = (1/\gamma)(v_t - v_u - \kappa\bar{v}(t-u) + \kappa\int v ds)$ from the CIR integral. (4) Sample $\int \sqrt{v} dW \wedge S | \int v ds \sim N(0, \int v ds)$, then assemble S_t . Step 2 is solved via the known closed-form cf of the integrated variance (a ratio of modified Bessel functions); invert it to the CDF F — originally by trapezoidal Fourier inversion, *today by COS* — and sample by inversion $\int v ds = F^{-1}(U)$, $U \sim \text{Uniform}(0,1)$.
15. Name the three tools behind the cf of the integrated variance (infinitesimal generator, time change, Girsanov) and say what discrepancy each one fixes relative to the squared Bessel process. **A.** (i) Infinitesimal generator — the CIR generator $\kappa(\bar{v}-x)f' + \frac{1}{2}\gamma^2 x f''$ is matched against the squared-Bessel generator $nf' + 2xf''$, whose Laplace transform of the time-integral is known (Pitman-Yor). (ii) Time change $s = \gamma^2 t/4$ rescales the *second-order* coefficient $\frac{1}{2}\gamma^2 x$ onto the squared-Bessel value $2x$. (iii) Girsanov change of measure absorbs the remaining *drift* discrepancy ($\kappa(\bar{v}-x)$ vs n), expressing the CIR integrated-variance transform as a ratio of squared-Bessel transforms — giving the Bessel-ratio formula for the cf.
16. Give the qualitative effect on the implied-vol surface of each parameter: γ , ρ , κ , ν_0 , $\bar{\nu}$. **A.** γ (vol-of-vol): larger γ deepens smile *curvature*. ρ : more negative ρ steepens the *skew* (heavier left tail). κ : small effect on smile shape; controls how fast ATM vol converges to $\sqrt{\bar{\nu}}$ in maturity. ν_0 : dominates *short*-maturity level. $\bar{\nu}$: dominates *long*-maturity level (both mediated by κ). Overall Heston gives heavier log-return tails than BS and reproduces the smile.

Lecture 08

- Define the Poisson process N_t with intensity λ . State its three defining properties and give $P(N_t = k)$. What are its mean and variance? **A.** N_t counts events with intensity λ : (i) $N_0 = 0$; (ii) independent, stationary increments; (iii) $N_t - N_s \sim \text{Poisson}(\lambda(t-s))$. So $P(N_t = k) = (\lambda t)^k e^{-\lambda t} / k!$, with mean = variance = λt .
- Show that mean = variance = λ for the Poisson law. Do the $E[X]$ sum by hand. **A.** $E[X] = \sum_{k \geq 0} k \lambda^k e^{-\lambda} / k! = \lambda e^{-\lambda} \sum_{k \geq 1} \lambda^{k-1} / (k-1)! = \lambda e^{-\lambda} e^{\lambda} = \lambda$. Computing $E[X(X-1)] = \lambda^2$ similarly gives $\text{Var}(X) = E[X^2] - E[X]^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda$. So mean = variance = λ .
- What is the compensated Poisson process, and why is it (not N_t itself) a martingale? Verify the martingale property explicitly. **A.** N_t has positive drift ($E[N_t] = \lambda t$), so it is not a martingale. The compensated process $\tilde{N}_t = N_t - \lambda t$ is. Check: $E[\tilde{N}_{t+\Delta} | \mathcal{F}_t] = \tilde{N}_t + E[N_{t+\Delta} - N_t] - \lambda \Delta = \tilde{N}_t + \lambda \Delta - \lambda \Delta$ (increment independent of $\mathcal{F}_t$, mean $\lambda \Delta$), and integrability holds. The compensator λt is the predictable finite-variation part removed to leave a martingale — the same idea as the risk-neutral jump drift correction.
- Distinguish homogeneous, inhomogeneous (time-dependent $\lambda(t)$), and Cox/doubly-stochastic Poisson processes. What is the compensator in each case? **A.** Homogeneous: constant λ , compensator λt . Inhomogeneous: deterministic $\lambda(t)$, counts Poisson with integrated intensity $\Lambda(a,b) = \int_a^b \lambda(s) ds$, compensator $\int_0^t \lambda(s) ds$. Cox/doubly-stochastic: $\lambda(t)$ itself a stochastic process (conditionally inhomogeneous given its path), compensator $\int_0^t \lambda(s) ds$. In all three $N_t - \int_0^t \lambda(s) ds$ is a martingale; the Cox case is the AJD slot $\lambda(x) = l_0 + l_1 \cdot x$ where intensity responds to the state.
- Derive the characteristic function of a compound Poisson process $Q_t = \sum_{k=1}^{N_t} Y_k$ by conditioning on N_t . You should reach $\exp(t\lambda(\hat{\nu}(u)-1))$. **A.** Condition on the count: $E[e^{iuQ_t}] = \sum_n P(N_t = n) E[e^{iu \sum_{k=1}^n Y_k}] = \sum_n ((\lambda t)^n e^{-\lambda t} / n!) \hat{\nu}(u)^n$ (Y_k i.i.d., $\hat{\nu}(u) = E[e^{iuY}]$). Pull out $e^{-\lambda t}$ and recognise the exponential series: $= e^{-\lambda t} e^{\lambda t \hat{\nu}(u)} = \exp(t\lambda(\hat{\nu}(u)-1))$.

6. Identify the characteristic exponent of the compound Poisson process and connect it to the Levy-Khintchine representation. Why does the “-1” appear, and what condition does $\nu_{\text{hat}}(0)=1$ enforce? **A.** The exponent is linear in t , so the characteristic exponent is $\psi(u)=\lambda(\hat{\nu}(u)-1)=\lambda\int(e^{\{iuy\}}-1)d\nu(y)$ — the finite-activity special case of Lévy-Khintchine $\psi(u)=\int(e^{\{iuy\}}-1)\Pi(dy)$ with Lévy measure $\Pi=\lambda\nu$. The “-1” is the compensation that makes $\psi(0)=0$, so $\varphi_{\{Q_t\}}(0)=1$ as any cf must satisfy ($\hat{\nu}(0)=1$ just says ν is a probability measure).
7. Write the Merton jump-diffusion SDE for S_t and derive the log-price SDE via Ito-for-jumps. Why is there no $(1/2)g''J^2$ term in the jump part, and why does $\ln S_t - \ln S_{\{t^-\}} = \ln(1+J_t)$? **A.** $dS_t = \mu S_{\{t^-\}} dt + \sigma S_{\{t^-\}} dW_t + J_t S_{\{t^-\}} dN_t$, so at a jump $S_t = S_{\{t^-\}}(1+J_t)$. Itô-for-jumps on $g = \ln$: $d \ln S = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW + [\ln S_t - \ln S_{\{t^-\}}]dN_t$. There is no $\frac{1}{2}g''J^2$ term because a jump is a *finite* move — we use the exact difference $g(Y_{\{t^-\}+J} - g(Y_{\{t^-\}}))$, not a second-order Taylor expansion (the $\frac{1}{2}g''$ term only handles the *continuous* Brownian part). And $\ln S_t - \ln S_{\{t^-\}} = \ln(S_{\{t^-\}}(1+J_t)) - \ln S_{\{t^-\}} = \ln(1+J_t)$.
8. Derive the risk-neutral drift correction $\mu = r - \lambda E[J_t]$ from the requirement that $e^{\{-rt\}}S_t$ is a Q-martingale. **A.** $d(e^{\{-rt\}}S_t) = e^{\{-rt\}}[(\mu - r)S_{\{t^-\}}dt + \sigma S_{\{t^-\}}dW_t + J_t S_{\{t^-\}}dN_t]$. The dW term is a martingale; taking E^Q and using $E[dN_t] = \lambda dt$ and $J \perp N$ gives drift $e^{\{-rt\}}S_{\{t^-\}}[(\mu - r) + \lambda E^Q[J]]dt$. Setting it to zero: $\mu = r - \lambda E^Q[J]$. The jump compensator $\lambda E[J]$ is the mean rate at which jumps push the price up; subtracting it keeps the expected return at r .
9. Common trap: in the Merton drift correction, is it $E[J]$ or $E[\ln(1+J)]$ that gets compensated? If $\ln(1+J) \sim N(\mu_J, \sigma_J^2)$, what is $\kappa = E[J]$? **A.** It is $E[J]$ (the arithmetic multiplier $J = S/S_{\{t^-\}} - 1$), *not* $E[\ln(1+J)]$. With $\ln(1+J) = \xi \sim N(\mu_J, \sigma_J^2)$, $1+J = e^\xi$, so $\kappa = E[J] = E[e^\xi] - 1 = e^{\{\mu_J + \frac{1}{2}\sigma_J^2\}} - 1$. The compensator is $\lambda\kappa = \lambda(e^{\{\mu_J + \frac{1}{2}\sigma_J^2\}} - 1)$. Confusing $E[J]$ with $E[\xi] = \mu_J$ is the classic exponent error.
10. Write Merton’s lognormal-jump cf of $\ln S_T$ and Kou’s double-exponential cf. State Kou’s $\nu_{\text{hat}}(u)$ and explain the roles of p, q, η_1, η_2 and why $\eta_1 > 1$ is required. **A.** Merton ($\hat{\nu}(u) = e^{\{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2\}}$, $\kappa = e^{\{\mu_J + \frac{1}{2}\sigma_J^2\}} - 1$): $\varphi(u) = \exp(iu(x + (r - \frac{1}{2}\sigma^2 - \lambda\kappa)\tau) - \frac{1}{2}\sigma^2 u^2 \tau + \lambda\tau(e^{\{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2\}} - 1))$. Kou: up-jumps $\text{Exp}(\eta_1)$ w.p. p , down-jumps $-\text{Exp}(\eta_2)$ w.p. $q = 1 - p$, so $\hat{\nu}(u) = p \eta_1 / (\eta_1 - iu) + q \eta_2 / (\eta_2 + iu)$, and $\varphi(u) = \exp(iu(x + (r - \frac{1}{2}\sigma^2 - \lambda\kappa)\tau) - \frac{1}{2}\sigma^2 u^2 \tau + \lambda\tau(\hat{\nu}(u) - 1))$. η_1 governs the up-tail decay, η_2 the down-tail, p/q the up/down probabilities; $\eta_1 > 1$ is required so $E[e^X] = E[1+J]$ is finite (the jump multiplier has finite mean, κ finite).
11. Compare Merton vs Kou: tail shape, symmetry, and the decay rate of the cf in u . Why does the cf decay rate matter for COS? **A.** Merton: Gaussian (symmetric, smooth) jumps; cf decays super-exponentially like $e^{\{-cu^2\}}$. Kou: exponential tails (heavier, asymmetric), and the cf decays only rationally like $1/u^2$. The cf decay rate sets the decay of the COS cosine coefficients, hence the number of terms N for a given accuracy: Merton needs few terms, Kou needs more; Kou also fattens the tails so the cumulant range $[a, b]$ is wider.
12. How does the jump-diffusion cf enter the COS formula? At which arguments is ϕ evaluated, and what is the ONLY model-specific input COS needs? **A.** The cf is the only model-specific input: $v \approx e^{\{-r\tau\}} \sum_k \text{Re}\{\phi(k\pi/(b-a))e^{\{-ik\pi a/(b-a)\}}\} V_k$, with ϕ evaluated at the grid arguments $u_k = k\pi/(b-a)$ (and spot carried as $\varphi(u)e^{\{iux_0\}}$). Nothing else about the jump structure is needed — supply Merton/Kou ϕ and the European COS machinery is unchanged (slide 32: COS works as long as the cf is known).
13. Short-maturity COS trap: write c_1, c_2, c_4 for Merton’s log-price. Why can a range $[a, b]$ built from c_2 alone be too narrow at small τ , and how does the jump c_4 fix it? (Connect to the Assignment 2 bug.) **A.** Merton cumulants: $c_1 = (r - \frac{1}{2}\sigma^2 - \lambda\kappa)\tau + \lambda\tau\mu_J$, $c_2 = \sigma^2\tau + \lambda\tau(\mu_J^2 + \sigma_J^2)$, $c_4 = \lambda\tau(\mu_J^4 + 6\mu_J^2\sigma_J^2 + 3\sigma_J^4)$. At small τ the diffusion part of c_2 ($=\sigma^2\tau$) shrinks, but jumps can still dominate the tails; a range built from c_2 alone (forgetting the jump kurtosis c_4 , or taking L too small) is too narrow, so the truncation-error floor stays high and the price is wrong *no matter how many terms N* — exactly the Assignment-2 bug. Keeping the $\sqrt{c_4}$ term (whose pure-jump contribution fattens the tails) and a generous L widens $[a, b]$ correctly.
14. State the PIDE for a jump-diffusion option value. Which term is non-local, and why does this motivate Fourier methods over a PIDE grid solve? **A.** $V_t + (r - \lambda E[J])S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV + \lambda E^Q[V(S(1+J), t)]$. The last term — an integral of V against the jump law — is *non-local*: it couples the value at S to values at all $S(1+J)$. On a grid this is a dense coupling (expensive); a transform method instead reads ϕ once and integrates against the payoff, so Fourier/COS sidesteps the non-local PIDE entirely.

15. Define an affine process and write the AJD SDE with all four affine conditions (μ , σ , σ^T , $\lambda(x)$, $R(x)$). What is the jump transform $\theta(c)$? **A.** Affine: the conditional cf is exponential-affine, $E[e^{\{u \cdot X_T\}} | X_t = x] = e^{\{\alpha + \beta \cdot x\}}$. The AJD is $dX = \mu(X)dt + \sigma(X)dW + dZ$ with all coefficients affine: drift $\mu(x) = K_0 + K_1 x$; covariance $(\sigma \sigma^T)\{ij\} = (H_0)\{ij\} + (H_1)\{ij\} \cdot x$; jump intensity $\lambda(x) = l_0 + l_1 x$ (with jumps $J \sim \nu$); short rate $R(x) = \rho_0 + \rho_1 x$. The jump law enters only through the jump transform $\theta(c) = \int e^{\{c \cdot z\}} d\nu(z)$ (the $\hat{\nu}$ with complex argument $c = \beta$).
16. State the AJD result $\phi = \exp(\alpha + \beta \cdot x)$ and the Riccati ODEs for (α, β) with boundary conditions. Which term is quadratic, and where does $(\theta(\beta) - 1)$ come from? **A.** $\phi = e^{\{\alpha + \beta \cdot x\}}$ with $\beta = \rho_1 - K_1^T \beta - \frac{1}{2} \beta^T H_1 \beta - l_1 (\theta(\beta) - 1)$, $\alpha = \rho_0 - K_0 \cdot \beta - \frac{1}{2} \beta^T H_0 \beta - l_0 (\theta(\beta) - 1)$, boundary $\beta(T, T) = u$, $\alpha(T, T) = 0$. The β equation is a Riccati ODE — the *quadratic* term is the $\frac{1}{2} \beta^T H_1 \beta$ diffusion term. The $(\theta(\beta) - 1)$ comes from the expected jump contribution: $E[(\phi(X+J) - \phi(X))dN] = \phi(\theta(\beta) - 1)\lambda(x)dt$, the compensator-type term, identical to the $(\hat{\nu} - 1)$ of the compound-Poisson cf.
17. Recover the Black-Scholes log-price cf by solving the AJD ODEs with no jumps ($X = \ln S$). Show $\beta = iu$ and the α you obtain. **A.** No jumps ($l_0 = l_1 = 0$), $X = \ln S$. The ODEs give $\beta = 0$ with $\beta(0) = iu$, so $\beta \equiv iu$, and $\alpha = (r - \frac{1}{2}\sigma^2)\beta + \frac{1}{2}\sigma^2\beta^2 - r$. With $\beta = iu$: $\alpha(u, \tau) = (r - \frac{1}{2}\sigma^2)iu\tau - \frac{1}{2}\sigma^2 u^2 \tau - r\tau$. Hence $\phi(u, \tau) = \exp((r - \frac{1}{2}\sigma^2)iu\tau - \frac{1}{2}\sigma^2 u^2 \tau - r\tau + iuX)$ — the GBM log-price cf, recovered as the degenerate AJD. Adding $l_0(\theta(iu) - 1)$ reproduces the Merton cf.
18. The $(\theta - 1)$ / compensator motif appears at three scales in this lecture (compensated Poisson, drift correction, AJD ODE). State all three and explain why they are “the same idea”. **A.** (i) Compensated Poisson: $N_t - \lambda t$. (ii) Risk-neutral drift correction: $\mu = r - \lambda E[J]$. (iii) AJD ODEs: the $-l(\theta(\beta) - 1)$ term. All three subtract the *predictable mean rate of the jumps* (the compensator) so that the relevant object — the counting process, the discounted stock, the transform’s drift — becomes a martingale. They are the single idea “remove the predictable jump drift” at three scales.

Cluster 5 — American options

Lecture 09

1. Define the American early-exercise problem as an optimal-stopping problem. Write $v(S, t) = \max(\Lambda, \text{continuation})$ and $v_0 = \sup_{\tau} E^Q[e^{-r\tau} \Lambda(S_{\tau}, \tau)]$. What makes τ^* “free”? **A.** At each instant the holder takes the larger of exercising now and holding: $v(S, t) = \max\{\Lambda(S, t), \text{continuation}\}$, where continuation is the discounted risk-neutral expectation of v an instant later. Globally $v_0 = \sup_{\tau \in T_{[0, T]}} E^Q[e^{-r\tau} \Lambda(S_{\tau}, \tau) | S_0]$, the supremum over all stopping times. τ^* is “free” because the optimal stopping rule is not given in advance — it is itself an output of the problem, determined by where continuation falls to intrinsic.
2. What is the free boundary $S(t)$? *Characterise the continuation region ($v > \Lambda$, BS PDE holds) vs the stopping region ($v = \Lambda$). State the smooth-pasting conditions at $S(t)$ for the put.* **A.** $S(t)$ is the optimal exercise boundary, unknown a priori (hence “free”), separating the continuation region C (hold, $v > \Lambda$, where v satisfies the BS PDE with equality) from the stopping region S (exercise, $v = \Lambda$). For the put: $S < S(t) \Rightarrow \text{exercise}$, $v = K - S$; $S > S(t) \Rightarrow \text{hold}$, $v > K - S$. At the boundary value and slope match continuously — the smooth-pasting conditions $v(S, t) = K - S^*$, $\partial_S v(S^*, t) = -1$.
3. PROVE that an American call on a non-dividend stock equals the European call. Give BOTH the slide dominating-strategy argument and the parity lower-bound $C^{\text{eur}} = S - Ke^{-r(T-t)} + P^{\text{eur}} > S - K$. **A.** Dominating strategy: instead of exercising at t for $S_t - K$, short the stock (receive S_t) and keep the call; at T close the short paying $\min(K, S_T) \leq K$. The alternative gives S_t now and $\leq K$ later, which strictly dominates $S_t - K$ now once interest on the deferred K is counted — so early exercise is dominated at every $t < T$, hence worthless. Lower-bound version: by parity and $P^{\text{eur}} \geq 0$, $C^{\text{eur}} = S - Ke^{-r(T-t)} + P^{\text{eur}} \geq S - Ke^{-r(T-t)} > S - K$ for $r > 0$. The live call exceeds its intrinsic, so the early-exercise right is never used: $C^{\text{am}} = C^{\text{eur}}$.
4. Why does the same argument FAIL for the put, so early exercise can be optimal? Use parity $P^{\text{eur}} = Ke^{-r(T-t)} - S + C^{\text{eur}}$ and $Ke^{-r(T-t)} < K$. What is the economic intuition? **A.** Parity gives $P^{\text{eur}} = Ke^{-r(T-t)} - S + C^{\text{eur}}$, and $Ke^{-r(T-t)} < K$, so the live put can be worth less than its intrinsic $K - S$. When that happens, exercising now and banking the full K to earn interest beats holding — early exercise is strictly optimal, so there is no European-American

equivalence. Intuition: the put holder *wants the strike cash K early* to earn interest, whereas the call holder wants to *defer paying K* — the asymmetry is entirely the sign of the strike cash flow.

5. Why does delta-hedging make the European BS relation an EQUALITY but the American put only an INEQUALITY? Walk through both arbitrage directions and explain why only one survives. **A.** For a European option the delta-hedged position is perfectly riskless, so it must earn r exactly — equality. For the American put, hedging does not remove the holder's early-exercise right. If the hedged option *out-earns* cash, an arbitrageur longs it (owning the exercise right) and locks the surplus — a genuine arbitrage, competed away. If it *under-earns* cash, arbitraging would require *shorting* the put, but that hands the exercise decision to the counterparty, who can exercise during $[t, t+\Delta t]$ — no riskless profit. Only the second case survives, giving $LP_{Am} \leq 0$, hence an inequality.
6. Write the LCP for the American put: $(v-\Lambda) \geq 0$, $Lv \leq 0$, $(v-\Lambda) \cdot Lv = 0$. Interpret each line, why it equals an obstacle problem, and state terminal/boundary conditions. **A.** $(v-\Lambda) \geq 0$ (value dominates payoff, else buy-and-exercise arbitrage), $Lv \leq 0$ (the BS operator is a supersolution, from the hedging inequality), $(v-\Lambda) \cdot Lv = 0$ (complementarity: at each point one holds with equality). In C , $v > \Lambda \Rightarrow Lv = 0$ (BS PDE holds); in S , $v = \Lambda \Rightarrow Lv \leq 0$ (payoff is not a BS solution, exercise optimal). It is an obstacle problem because v is the smallest supersolution of L lying above the obstacle Λ . Conditions: terminal $v(S, T) = \max(K - S, 0)$; boundary $v(0, t) = K$, $v(\infty, t) = 0$.
7. State the Bermudan backward recursion in log-space: $c(x, t_{n-1}) = e^{-r\Delta t} E[v(X(t_n)) | x]$, $v = \max(\Lambda, c)$. What single quantity do all four methods compute differently? **A.** With $X(t_n) = \ln(S(t_n)/K)$, $\Lambda(x, t) = \max(\alpha K(e^x - 1), 0)$, the recursion for $n = N, \dots, 2$ is $c(x, t_{n-1}) = e^{-r\Delta t} E[v(X(t_n)) | x]$, $v(x, t_{n-1}) = \max(\Lambda(x, t_{n-1}), c(x, t_{n-1}))$. All four methods (tree, FD/LCP, LSM, COS) run exactly this Bellman recursion; they differ *only* in how they compute the conditional expectation c — the continuation value.
8. Binomial tree: write the backward step with $\max(\text{intrinsic}, \text{discounted continuation})$. What is the *ONLY* line that differs from a European tree? How does $p = (e^{r\Delta t} - d)/(u - d)$ arise? **A.** $V_m^{n=\max(K-S_m, n)} = e^{-r\Delta t} [pV_{m+1}^{n+1} + (1-p)V_m^{n+1}]$. The conditional expectation is the two-point binomial average; the *only* difference from a European tree is the outer $\max(\text{intrinsic}, \cdot)$ inserted at every node. $p = (e^{r\Delta t} - d)/(u - d)$ arises from the risk-neutral / martingale condition $E^{Q[S_{n+1} | S_n]} = e^{r\Delta t} S_n$, i.e. $pu + (1-p)d = e^{r\Delta t}$.
9. Finite-difference LCP: write the projected update. Why can't you just solve the implicit linear system and clip once? Explain PSOR — projecting *INSIDE* each SOR sweep. **A.** Projected update: $V_m^{n=\max(K-S_m, n)} = g(V_{m+1}^{n+1}, V_m^{n+1}, V_{m-1}^{n+1})$ with g the implicit BS stencil — the discrete LCP. For an *implicit* scheme the node m couples to its neighbours at the *same* time level through a tridiagonal system, and the constraint $V \geq \Lambda$ must hold simultaneously with that solve; solving the unconstrained system and clipping once gives the wrong answer because the clipped values feed back into the coupling. PSOR (Projected SOR) runs SOR sweeps but, after each node's update, immediately projects $V_m \leftarrow \max(\Lambda_m, V_m^{\text{SOR}})$; projecting *inside* the iteration converges to the true LCP solution.
10. LSM mechanics: explain the backward regression of discounted future cash flows on basis functions. Why regress only on ITM paths? Why is the per-path decision $\Lambda > \hat{c}$ (not the fitted value) the cash flow? **A.** Working backward, at each date regress each path's realised discounted future cash flow Y (from following the already-determined later policy, discounted to t_n) on basis functions of the current state $X = S_{t_n}$ (e.g. $1, X, X^2$); the fit $\hat{c}(x) = \sum \hat{\alpha}_j L_j(x)$ estimates the continuation value. Regress only on ITM paths because OTM paths have zero intrinsic and would never exercise — restricting fits the conditional expectation only where the decision matters (cheaper and more accurate). The realised *cash flow* used is the path's actual intrinsic when $\Lambda > \hat{c}$, not the fitted $\hat{c}$: $\hat{c}$ is only a noisy decision rule, while the realised payoff is the unbiased value collected on that path.
11. LSM 8-path example: how is Y constructed (which discount factor, which future date)? Why does the $t=1$ regression depend on the $t=2$ exercise decisions? Interpret 0.1144 vs European 0.0564. **A.** At $t=2$, Y is the $t=3$ payoff discounted one period ($\times e^{-0.06} = 0.94176$), $X = S_{t=2}$; fit gives exercise on paths 4, 6, 7. At $t=1$, Y is the *subsequent* realised cash flow — which by the $t=2$ decisions occurs at $t=2$ or $t=3$ — discounted back to $t=1$; so the $t=1$ regression depends on the $t=2$ exercise decisions because they determine when each path's

future cash flow actually lands. Final: discount each path's first-exercise cash flow to $t=0$ and average $\rightarrow$ American put 0.1144, vs European 0.0564 (only the $t=3$ payoffs). The American is $\sim$ twice the European: that gap is the entire value of the early-exercise right.

12. Is LSM biased? Explain why a suboptimal finite-basis policy gives a LOWER bound, and the competing in-sample foresight upward bias. How do you get a clean out-of-sample estimate? **A.** A finite basis cannot represent the true continuation surface exactly, so the estimated stopping policy is suboptimal and a suboptimal rule can only *lose* value — giving a lower bound. There is a competing *upward* (foresight) bias if the price is read off the same cash flows used to fit the regression (in-sample look-ahead). Clean practice: use the regression only for the exercise *decision* and value the realised cash flows along each path's stopping time; for a strictly out-of-sample estimate, freeze the policy and re-simulate fresh paths.
13. COS Bermudan: derive $c(x, t_{n-1}) \approx e^{-r\Delta t} \sum' \text{Re}\{\phi(k\pi/(b-a)) e^{ik\pi(x-a)/(b-a)}\} V_k(t_n)$. What ARE the $V_k(t_n)$ — relate to the Lecture-6 European V_k . **A.** Write $c(x, t_{n-1}) = e^{-r\Delta t} \int v(y, t_n) \bar{f}(y|x) dy$ — the European COS integral with the payoff replaced by the next-step value $v(\cdot, t_n)$. Inserting the cosine expansion of f (coefficients read off ϕ) gives $\hat{c}(x, t_{n-1}) = e^{-r\Delta t} \sum' \bar{f}_k \text{Re}\{\phi(k\pi/(b-a)) e^{ik\pi(x-a)/(b-a)}\} V_k(t_n)$. Here $V_k(t_n) = (2/(b-a)) \int_a^b v(y, t_n) \cos(k\pi(y-a)/(b-a)) dy$ are the cosine coefficients of the option *value* at t_n . At t_N they are exactly the Lecture-6 European payoff coefficients; for $n < N$ they are built recursively — so it is a backward recursion on the coefficient vector, not on a price grid.
14. COS early-exercise point: define x_n^* by $c(x_n, t_n) = \Lambda(x_n, t_n)$. Write the split $V_k(t_n) = G_k(a, x_n) + C_k(x_n, t_n)$ for a put. Which piece is closed-form (χ_k, ψ_k), which needs the previous coefficients? How is x_n^* found? **A.** x_n^* solves $c(x_n, t_n) = \Lambda(x_n, t_n)$ — the discrete avatar of the free boundary $S(t)$. For a put, exercise is on the low- y side: $V_k(t_n) = G_k(a, x_n) + C_k(x_n, b, t_n)$, where G_k is the cosine integral of the payoff $\Lambda = \alpha K(e^{-y} - 1)$ and C_k that of the continuation c . G_k is closed-form — a combination of the Lecture-6 χ_k (the e^{-y} cos integral) and ψ_k (the cos integral) on the cut interval. C_k needs the previous* step's coefficients $V_l(t_{n+1})$ (it is built from $\hat{c}$). x_n^* is found by Newton's method (one scalar root-find; the derivative of $\hat{c}$ is the differentiated cosine sum).
15. COS algorithm: initialisation, backward loop (Newton for x_n , FFT for C_k), reconstruction. Why is the cost $O(NK \log_2 K)$ not $O(NK^2)$? What gives the FFT (Toeplitz/Hankel kernel)? **A.** Initialise at t_N with the pure payoff coefficients $V_k(t_N) = G_k$ (Lecture-6 European). Backward loop $n = N-1, \dots, 1$: (a) find x_n by Newton from $\hat{c} = \Lambda$; (b) form $V_k(t_n)$ via the split — G_k closed-form, C_k by the FFT-accelerated product on $\{V_l(t_{n+1})\}$. Reconstruct: $v(x_0, t_0) = e^{-r\Delta t} \sum' \text{Re}\{\phi(k\pi/(b-a)) e^{ik\pi(x_0-a)/(b-a)}\} V_k(t_1)$. The C_k coupling matrix $M_{\{k,l\}}$ is Hankel-plus-Toeplitz (depends on $l \pm k$), so the matrix-vector product is a discrete convolution done by FFT in $O(K \log_2 K)$ instead of the naive $O(K^2)$; over N steps the cost is $O(NK \log_2 K)$.
16. Why does the cumulant range $[a, b] = c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}$ caveat carry into the Bermudan COS recursion and bite harder per-step? Tie to the Assignment-2 bug: $c_2 = \sigma^2 \Delta t \rightarrow 0$, $c_4 = 0$, no floor; adding K terms does NOT help. **A.** The range is still set from the cumulants of X_T , but now it must contain the support over the *whole* backward recursion and the Newton x_n^* must land inside $[a, b]$. At short maturity (or small per-step Δt) $c_2 = \sigma^2 \Delta t \rightarrow 0$, so the box collapses toward c_1 ; for a Gaussian model $c_4 = 0$ means no $\sqrt{c_4}$ floor — the same failure mode as the Assignment-2 bug. Adding cosine terms K cannot cure a range (truncation) error; the fix is a wider L or a minimum half-width floor. It bites per-step because each of the N steps re-uses the same possibly-too-narrow box.
17. Richardson extrapolation Bermudan $\rightarrow$ American: why does every method here price a Bermudan? Write the 4-point scheme and explain what it cancels. **A.** Every method discretises exercise to a finite monitoring set, so each prices a *Bermudan*; the American is the $N \rightarrow \infty$ limit. Rather than push N large, take Bermudan prices $v(M)$ at $M = 2^{d,2}\{d+1\}, 2^{d+2,2}\{d+3\}$ and combine: $v_{\text{Am}}(d) = (1/21)(64v(2^{\{d+3\}}) - 56v(2^{\{d+2\}}) + 14v(2^{\{d+1\}}) - v(2^d))$. The weighted combination cancels the leading error terms in the monitoring spacing, accelerating convergence to the continuous-American limit from a handful of cheap fast Bermudan computations.

Cluster 6 — Exotics

Lecture 11

1. Give the precise payoff of an arithmetic-average Asian call and a geometric-average Asian call.
A. Fixed-strike arithmetic-average call: $((1/n)\sum_{i=1}^n S_{t_i} - K)^+$. Geometric-average call: $((\prod_{i=1}^n S_{t_i})^{1/n} - K)^+$. (Floating-strike variants replace K by a multiple of S_T). Both are path-dependent on the monitoring dates $t_1, \dots, t_n$.
2. What economic need does an Asian option serve that a vanilla cannot, and why is the premium lower? (airline fuel hedge; average exposure; effective variance reduced) **A.** It matches an exposure to an *average* rather than a terminal price — e.g. an airline hedging a month of fuel procurement buys one Asian call on the monthly average instead of ~20 daily vanillas, matching its real cost $(1/n)\sum S_{t_i}$ and reducing fixing-manipulation risk. The premium is lower because the average is less volatile than the terminal price (continuous-monitoring variance $\sigma^2 T/3$, one-third of the vanilla), so the convex payoff is worth less.
3. Derive the geometric-Asian closed form: why is G lognormal, and what are μ_G and σ_G^2 for equally spaced dates? Show $\sigma_G^2 \rightarrow \sigma^2 T/3$ in the continuous limit. **A.** $\ln G = (1/n)\sum [(r - q - \frac{1}{2}\sigma^2)t_i + \sigma W_{t_i}]$ is an affine function of the jointly Gaussian $(W_{t_1}, \dots, W_{t_n})$, hence normal — so G is lognormal and prices by a Black-style formula. $\mu_G = \ln S_0 + (r - q - \frac{1}{2}\sigma^2)\bar{t}$, $\sigma_G^2 = (\sigma^2/n^2)\sum_{i,j} \min(t_i, t_j)$. For equally spaced $t_i = i\Delta t$: $\bar{t} = (n+1)T/2n$, $\sigma_G^2 = \sigma^2 \Delta t (n+1)(2n+1)/6n^2 \rightarrow \sigma^2 T/3$ as $n \rightarrow \infty$. The $\sigma^2 T/3$ (vs $\sigma^2 T$) is exactly why the Asian is cheaper, and the geometric Asian is the standard MC control variate for the arithmetic one (Kemna-Vorst).
4. Why does the arithmetic Asian have NO closed form / no closed-form CF, and what does that imply for method choice? (sum of lognormals not lognormal) **A.** $A_T = (1/n)\sum S_{t_i}$ is a *sum* of lognormals, which is not lognormal and has no elementary density — and crucially no closed-form characteristic function even under affine models. So one cannot price it by Black or apply COS directly to A_T ; the options are moment-matching, the (Vecer-reduced) PDE, COS with a cumulant-approximated/recursive cf, or Monte Carlo.
5. List the four numerical methods for arithmetic Asians and the one-line reason for each; what is the standard control variate for the MC? (geometric Asian) **A.** (i) Moment matching (Turnbull-Wakeman/Levy) — fit a lognormal to the first two moments of A_T ; fast but an approximation that degrades in the tails. (ii) PDE — 2-D in (S, A) , reduced exactly to 1-D by Vecer's similarity trick. (iii) COS/Fourier (Zhang-Oosterlee) — cumulant-approximated ϕ_{A_T} or backward recursion over monitoring dates with the one-step conditional cf. (iv) Monte Carlo — most flexible for path dependence, with the *geometric*-Asian closed form as control variate (essentially mandatory variance reduction).
6. State the Fouque-Han 3D Asian PDE and identify the four blocks. Then explain Vecer's reduction: what does $q(t)$ replicate, and how does it collapse the path-dependent state I_t ? **A.** With (S, Y, I) , $I_t = \int_0^t S_u du$, the floating-strike Asian solves $\partial_t P + \frac{1}{2}\sigma^2 S^2 \partial_{SS} P + \rho \beta f s \partial_{SY} P + \frac{1}{2}\beta^2 \partial_{YY} P + r(s - I/T)P = 0$ terminal $(s - I/T)^+$. Four blocks: second-order (s, y) diffusion (cross-term carries ρ); first-order risk-neutral drift in s and risk-adjusted drift in y ; the pure-drift term $s \partial_I P$ (since $dI = S dt$); and $-rP$ discounting. Vecer: hold a *time-deterministic* number of shares $q(t) = (1 - e^{-r(T-t)})/(rT)$ in a self-financing portfolio so that $X_T = (1/T)\int_0^T S_u du - K$ — i.e. $q(t)$ replicates the running average. Trading this deterministic q collapses the path-dependent state I_t into the single stochastic wealth X_t ; switching to the stock numéraire and using homogeneity reduces the 3-D PDE to 2-D (1-D in the constant-vol limit, the Vecer Asian PDE).
7. Give both digital payoffs (cash-or-nothing, asset-or-nothing). Derive the cash-or-nothing call price = $e^{-rT} N(d_2)$ from $Q(S_T > K)$. **A.** Cash-or-nothing call: $1_{\{S_T > K\}}$ ($\times$ notional Q). Asset-or-nothing call: $S_T 1_{\{S_T > K\}}$. The cash-or-nothing pays \$1 on $\{S_T > K\}$, so $C_{\text{dig}} = e^{-rT} Q(S_T > K)$. Under GBM, $\{S_T > K\} = \{Z > -d_2\}$ with $d_2 = (\ln(S_0/K) + (r - q - \frac{1}{2}\sigma^2)T)/(\sigma\sqrt{T})$, so $Q(S_T > K) = N(d_2)$ and $C_{\text{dig}} = e^{-rT} N(d_2)$. A digital prices a risk-neutral probability; the asset-or-nothing is $S_0 e^{-qT} N(d_1)$, and vanilla = asset-or-nothing - $K \cdot$ cash-or-nothing.
8. A digital has a clean closed form. Why is it nonetheless considered a hard/dangerous product? (delta spike, gamma sign-flip, pin risk; call-spread notional $1/\epsilon \rightarrow$ infinity near K at expiry) **A.** Valuation is trivial ($e^{-rT} N(d_2)$); the *hedge* is not. The replicating call spread has notional

$1/\varepsilon \rightarrow \infty$ as the spread tightens, and near expiry near the strike the delta spikes (Dirac-like) and gamma flips sign — “pin risk”. A desk delta-hedging a sold digital must trade a huge, sign-changing position in a vanishingly small neighbourhood of K as $t \rightarrow T$, impossible to execute cleanly; in practice they over-hedge by pricing/hedging a (super-replicating) call spread, not the exact digital. The difficulty is risk management, not pricing.

9. State the Breeden-Litzenberger link $dC/dK = -e^{-rT} Q(S_T > K)$ and explain the centred call-spread static replication (and why centring gives 2nd-order accuracy). **A.** $\partial C/\partial K = -e^{-rT} Q(S_T > K) = -C_{\text{dig}}$: a digital is minus the strike-derivative of vanilla prices, linking it to the implied (risk-neutral) distribution. Static replication: $1_{\{S_T > K\}} \approx [(S_T - K + \varepsilon/2)^+ - (S_T - K - \varepsilon/2)^+]/\varepsilon$, i.e. buy $1/\varepsilon$ calls at $K - \varepsilon/2$ and sell $1/\varepsilon$ at $K + \varepsilon/2$. *Centring* the spread symmetrically around K makes the leading $O(\varepsilon)$ error cancel (the approximation is a centred difference of the call price in K), giving second-order $O(\varepsilon^2)$ accuracy; a one-sided spread is only first-order.
10. Which numerical method is most natural for a digital and why? (COS: indicator payoff has closed-form cosine coefficients; it's the European COS method as a special case) **A.** COS. The indicator payoff has closed-form cosine coefficients H_k (an elementary cos-integral over the in-the-money half-range), so a digital is just the European COS method with a different H_k — no discontinuity penalty on the payoff side, since the coefficients are computed analytically rather than by sampling the jump. (Closed form exists under GBM; COS handles any model with a known cf.)
11. Give the basket and worst-of payoffs. Show $\text{Var}(B_T) = \sum_i \sum_j w_i w_j \text{Cov}(S_i, S_j)$ and explain why correlation is the dominant risk. **A.** Basket call: $(\sum_i w_i S_i(T) - K)^+$. Worst-of call: $(\min_i S_i(T) - K)^+$. For $B_T = \sum_i w_i S_i(T)$, $\text{Var}(B_T) = \sum_i \sum_j w_i w_j \text{Cov}(S_i, S_j) = \sum_i w_i^2 \text{Var}(S_i) + \sum_{i \neq j} w_i w_j \rho_{ij} \sqrt{(\text{Var } S_i \text{Var } S_j)}$. The off-diagonal ρ_{ij} terms drive the variance: positively correlated baskets are more volatile (dearer), weakly/negatively correlated ones diversify. A worst-of is even more correlation-sensitive ($\rho \rightarrow 1$ makes it a single-asset option; low ρ drags the minimum down). The 2008 correlation break-down is what made these books dangerous.
12. Why is Monte Carlo (not PDE or COS) the method for a $n \geq 4$ -asset worst-of? State the COS availability honestly. (curse of dimensionality; Ruijter-Oosterlee 2D only, no dedicated COS for $n \geq 4$) **A.** PDE and COS both use a tensor-product grid whose cost grows like M^n — the curse of dimensionality kills them beyond $n \approx 2-3$. Honestly: COS exists for 2-asset basket/call-on-min (Ruijter-Oosterlee 2012) and a 3-asset spread (Pellegrino-Sabino 2016), but there is *no dedicated COS method for $n \geq 4$* . MC handles arbitrary dimension and correlation naturally (Cholesky-correlate normals, simulate, evaluate $(\min_i S_i - K)^+$), enhanced with anti-thetics/QMC/control variates — hence the industry standard for the multi-asset book.
13. In a worst-of autocallable, is the embedded option a call-on-min or a worst-of put? Explain who is short what. **A.** A worst-of *put*, not a clean call-on-min. The investor is *short* downside on the weakest index (e.g. redemption is reduced if $\min_i S_i(T)/S_i(0)$ falls below a 60% barrier), implicitly *selling* correlation and downside risk; the high coupon funds that sold risk. The issuing bank is long that protection (short correlation in its book). Be ready to state that the traded structure is a worst-of put, distinct from the textbook call on the min.
14. Give the cliquet payoff with local cap C and floor F . Why is it forward-skew / forward-volatility sensitive, and why is MC with local/stochastic vol the practical method? **A.** With period returns $R_i = S_{\{t_i\}}/S_{\{t_{i-1}\}} - 1$, the cliquet pays $\sum_i \min(\max(R_i, F), C)$ (optionally with a global cap/floor on the sum). It is essentially a strip of *forward-starting* capped/floored options: each l_i is determined over the future window $[t_{i-1}, t_i]$, so its value depends on the *forward* volatility and forward skew prevailing then, not today's spot-starting smile — an exposure no vanilla portfolio (all spot-starting) can replicate. So pricing is highly sensitive to smile *dynamics*; the practical method is MC with a local/stochastic-vol (or hybrid) model that gets the forward skew right, since the reset features combine trivially with such models.
15. Pre-2008 vs post-2008: which exotics grew, which shrank, and what is the unifying post-crisis preference? (hedgeable, modellable, explainable, liquid; basket/worst-of and autocallables grew; bespoke/long-dated/aggressive cliquets shrank) **A.** Grew: basket/worst-of and autocallables (now dominant in equity structured products), driven by retail yield demand outrunning the correlation caveats. Shrank: highly bespoke OTC, long-dated path-dependent exotics, and aggressive cliquets (now insurance-driven, reduced). Barriers/Asians/digitals

stayed active. The unifying post-crisis preference is for products that are easier to *hedge, model, explain to regulators, and more liquid* — reflecting XVA/funding costs, Basel III capital, collateralisation, and the lessons of counterparty risk and correlation break-down.